



**Layout Alternative Exploration and Analysis of
the UAH SMAP Center's Outreach Program
Office**

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Executive Summary:

This study seeks to identify an alternative layout for the UAH SMAP Center Outreach Program's office space. While the existing office space is functional, there are numerous non-value-added activities along with other concerns which can be addressed through the introduction of a new layout. This study proposes a total of three alternative layouts which are derived from findings captured in a literature review, interviews with UAH SMAP Center employees, strategic documents, and preference elicitation from the governing stakeholders of the system. A substantial amount of data is collected and analyzed in order to assess each of the three alternative layouts. These layouts are then compared on the basis of several qualitative and quantitative criteria. In order to select the proper layout among the three options, a model was constructed to weight the various objectives expressed by the stakeholders. Additionally, simulation models (using Simio) were created to represent the operational performance of each layout.

This study ultimately concludes that the combined office layout (inspired by Lean principles uncovered in the literature review) would be the most suitable layout to address the governing stakeholder objectives. This study also provides a discussion about best practices for implementing a new layout of this type. A description of the qualitative and quantitative implications of incorporating the combined office layout is also a central focal point for discussion.

“An employee's thoughts about the broad subject of space layout are comparable to his thoughts about the air he breathes. Air is no problem as long as he experiences no difficulty in getting enough. But make it difficult for him to get enough air to breathe, and air becomes a problem of extreme importance.” --- Walter Waddel [3]

Introduction:

The UAH SMAP Center was established in 1988 to provide expertise, leadership, and support to the Army, NASA, other government agencies, and private sector organizations. A critical aspect of the UAH SMAP Center is the Outreach Program, which strives to adapt technology to achieve new capabilities which in turn is used to educate students around the state of Alabama on STEM topics. Since moving to its current location over ten years ago, SMAP has been organically evolving its layout in its' current space. Currently, the SMAP is dealing with limited space to expand into, unavailable areas for storage, and the significant time required to complete tasks. This study identifies the common cause of these issues to be the layout of the office space itself. Additionally, this study conducts a survey of the scientific studies and relevant background literature on the topic and develops a total of three layout alternatives consisting of the Open Office, Systems Furniture (cubicle), and Combined (lean principles). Analysis is conducted to compare the three alternatives with a quantifiable characteristic of the layout's effect linear distance traveled (in hours paid to employees) to complete the center's current task load. The expected qualitative results of each configuration are also compared as informed by the current literature and relevant scientific studies. Ultimately, this study finds that the Combined Layout is the ideal choice as it is able to double the efficiency of operations from the current state (even though it was not as good as the other two layouts in this regard), while also having the least change made to the system (which is consistent with what the qualitative measures and scientific literature pointed to). More broadly speaking, this study concludes that office layouts, in practice, are systems with a complex series of interactions wherein small changes can have significant counterintuitive and unpredictable outcomes. Because of this, this study recommends the SMAP

center to mitigate this risk before making any drastic change to their operational layout beyond the changes recommended.

1. Recognizing the Problem:

The Outreach Program hosted within the University of Alabama in Huntsville (UAH) Systems Management and Production Center (SMAP Center) strives to serve the north Alabama community through outreach to high schools. These outreach activities are fundamentally based upon STEM education initiatives to enhance technology awareness and enthusiasm for STEM disciplines in many of the underfunded and disadvantaged high schools in north Alabama. Unlike other STEM outreach programs, the UAH SMAP Center strives to incorporate an innovative twist to the ongoing work activities contained under the purview of the program. Specifically, employees associated with the program make advancements on projects that seek to adapt technology to fit new needs and purposes. These new technological applications can come in a wide variety of forms ranging from nursing applications, facility safety applications, military defense applications, science research applications, and much more. These advancements favorably compliment the outreach function of the program because these technological improvements may be leveraged to further engage and capture the interest of the next generation of scientists and engineers. The functions of the outreach program described here also relate to the overarching goals of the UAH SMAP Center as a whole.

From an operations perspective, the program operates on the basis of student workers who are compensated at an average rate of \$12.00 per hour. While some variation in wages exists due to differences in experience, these deviations are minimal and the job duties are not influenced by

small changes in wages within this program. In total, there are 18 part-time employees that work on a part time basis. These employees are responsible for fulfilling the daily activities of the program which are strategically tailored to address the overarching purpose and goals of the program at large. Each employee holds a particular office space within the layout of the system of interest.

Presently, the outreach program at the UAH SMAP Center is facing several prominent issues. These include an inflated amount of time required to complete tasks, a lack of available areas for storing resources, and having a limited amount of space to expand into. Overall, these issues most strongly impact the long-term success and growth of the program at hand. While the program is still able to function on a daily basis, the natural growth of the program is stunted by the operational constraints imposed on the program. Currently, there is no office space available to house new workers. From an organizational perspective, labor costs have become somewhat inflated due to the minimal return per dollar as dictated by the tremendous inefficiencies and operational constraints which are imposed on the system at hand.

Upon examining the system at hand, it was apparent that an element of commonality was present among all of these issues. This driving force appears to be the layout and arrangement of the resources within the system. It is hypothesized that altering the layout to eliminate the inherent inefficiencies would effectively address the issues previously mentioned that the program is experiencing. Over the years, the outreach program has naturally growth and expanded their scope of operations. As such, the existing layout is an emergent property of the accumulation of reactionary measures to address various issues. There have never been any intentional design activities that have attempted to optimize this layout or the arrangement of the equipment. Due to the highly ideal location of the UAH SMAP Center, moving is a highly unlikely outcome. Thus,

improving the existing state of the system on the basis of the physical layout and arrangement of resources is critical to long-term success.

2. Define the Objective:

The overarching goals of the study have been clearly defined in accordance with the stakeholder desires for system performance. As always, it is desired to achieve a state of operational excellence that captures all of the benefits associated with an efficient layout while avoiding the negative impacts related to such a change. However, this idealized conception of the system is not always possible. For office environments in particular, there are numerous qualitative considerations that must be acknowledged that would not be relevant in a traditional manufacturing context. Thus, this study seeks to capture the qualitative implications of adaptations to a layout while also strongly pursuing various quantitative metrics that provided valuable insight to the efficiency, effectiveness, and suitability of the layout in a variety of regards. Given this situation, a compiled list of the stated objectives under the purview and scope of this study are provided below.

1. *The Working Environment and the Office Culture Must be Accurately Accounted for.* This aspect will be largely captured through the effective consideration and analysis of the qualitative impacts of various layouts. The specific culture that is present in any operating environment may be difficult to capture depending on the circumstances. However, this study seeks to make noteworthy improvements to the layout without imposing negative drawbacks on the work culture or operational standards [26] [27] [28] [29] [30].

2. *Increase the Productivity of the UAH SMAP Center Outreach Program.* This study seeks to increase the productivity of the outreach program by altering the layout that houses the program. In effect, this will reduce the non-value-added motions that are involved and required for the daily operations while not infringing upon the qualitative considerations which also positively contribute to productivity levels. Overall, this study shares some of the same motivational undertones that are captured in the popularized Lean methodologies for enhancing operational efficiencies.
3. *The Solution Must be Implementable.* While there are a wide range of feasible solutions to the problem at hand that would yield beneficial results, some of these would be incredibly difficult to implement, thereby diminishing the value of such a solution. It is desired to utilize the resources that are currently available within the system to identify positive change which would bypass the inefficiencies that the current state of the system is presently experiencing.

3. Assemble the Relevant Data:

The collection of data for usage under the scope of this study was primary comprised of two components. First, a survey of the existing literature relating to work design theory, office layout practices, and Lean principles was conducted using scientific journals and published literature. Second, extensive activities were conducted to capture as much data as possible about the current state of the system at hand in order to effectively and accurately characterize the state of operations. The findings from both of these efforts will be utilized as the basis of analysis for the study going forward.

Upon surveying the dimensions of each room and identifying the locations of various areas of access within the system, a layout was generated to depict the current state. The results of this are shown in Figure 1. Additionally, extensive data collection activities were conducted to determine how often each area of access was utilized by the employees housed within the system. While some level of variability is naturally present in the quantitative data captured, it was determined that a deterministic approach was suitable for the analysis tools at hand. When needed, a stochastic representation of the data will be used to provide further insights. The results of these investigations are depicted in Figure 2.

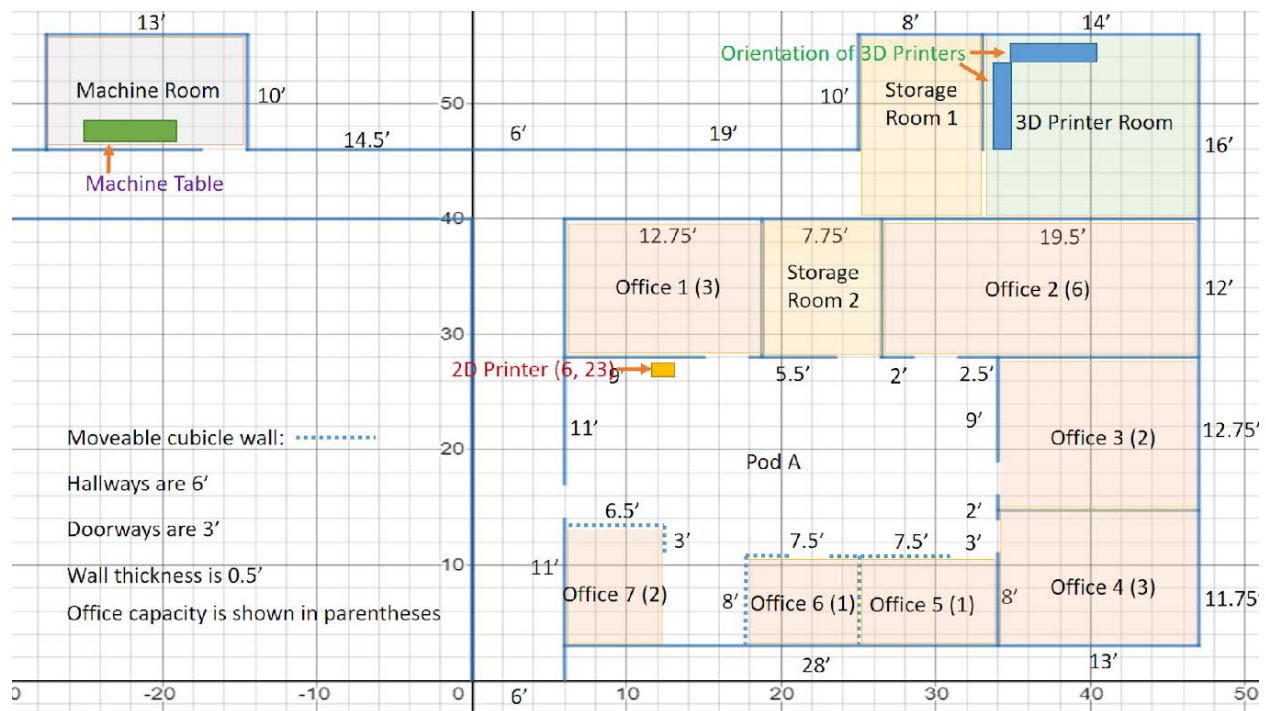


Figure 1. Layout of the Current State of the System.

	3D Printers	Storage (Total)	2D Printer	Machines
Demand	41 accesses/day	20 accesses/day	16 accesses/day	3 accesses/day

Figure 2. Total Demand Associated with Each Workstation.

Within the diagram shown above, each room and area of access plays a critical role in the operational function of the systems on a daily basis. In the upper right region of the diagram, the 3D printer room is depicted. The purpose of this room (in the current state of operations) is to house the 3D printers along with a handful of associated tools and supplies that may be relevant to the sustainment of operations for the various 3D printers. Next to the 3D printer room is the first of two storage rooms. These storage rooms house a variety of items for both long-term and short-term storage purposes. Hardware components, cleaning supplies, 3D printer supplies, electrical component, and various tools are commonly stored in these storage rooms. In the upper left region of the diagram, the machine room is depicted. While small in size, this room can hold a few different types of machines. Even though there is some variety in the machines available, these resources are not extensive enough to be regarded as a production facility. Also, the variety of machinery available is mostly only capable to offering single operations on a particular part when occasionally needed. Thus, the machine room is not a frequently accessed area, but it does offer some appealing capabilities to address situational circumstances associated with various projects. Aside from this, offices are featured in the diagram. As currently defined, there are a total of seven offices that hold varying amounts of people. Some offices are larger than others in order to enable higher levels of collaboration on projects. Despite the differences in positioning for various offices, the common area is denoted as “Pod A.” This area is multifunctional and may be used for presentations and meetings when needed. Most importantly, this space is the most

accepting and flexible to layout changes. Furthermore, this broad communal area hosts a singular 2D printer which is a common area of access for the students housed in the associated offices. In regards to dimensions, the 3D printers possess dimensions of 1.75 feet wide by 1.5 feet deep. Likewise, the machines hosted in the machine room are arranged on a 7 foot by 2 foot table and they should remain oriented in this fashion for optimal storage. The single 2D printer presented in Pod A has dimensions which are 1.75 feet wide by 1.5 feet deep (which is the same physical footprint as the 3D printers). A few movable cubicle walls are depicted in the diagram. These walls are subject to change in accordance with the demands of each alternative layout that is proposed by the study. For the initial state, it has been determined (based on the dimensions) that the average office space per person is about 51.33 ft^2 , but this value is expected to fluctuate depending on the alternative layouts considered.

In order to establish a foundation for quantitative analysis, a variety of data was collected over numerous days of observing the operation of the system at hand. This data, in conjunction with research, was utilized to formulate a deterministic representation of the speed that a typical student walks to access various locations within the system. An average rate of $4.43 \frac{\text{feet}}{\text{second}}$ is applied under the governing formula $D = R \cdot T$ to model the non-value-added time required for various trips to workstations [15] [16] [17]. The roundtrip time associated with accessing a particular area is found by doubling the time required to make a one-way trip to that location. From a broad perspective, the generalized relationships used for this study are depicted in the equations below.

$$\frac{\text{Distance}}{\text{Walking Speed}} = \frac{\text{Distance}}{4.43 \frac{\text{feet}}{\text{second}}} = \text{Time} = 0.5 \cdot \text{Roundtrip Time}$$

When considering the hourly rate of the student workers within the system at hand, these governing equations can be related to a cost factor that expresses the non-value-added time that is spend on unnecessary movement as an approximate dollar value. Despite the relatively low wage rate and the part-time status of the students contained in the system, this will be utilized to express cost savings. The resulting expression for cost is depicted in the generalized relationships shown below.

$$(Hourly\ Rate) \cdot RT\ Time = \frac{\$12}{1\ hour} \cdot RT\ Time = \frac{\$12}{3600\ seconds} \cdot RT\ Time = RT\ Cost$$

Expanding upon this work, a weighted average cost will be found that accounts for the number of employees in each office. Some offices contain more employees than other offices, so being able to capture a proper depiction of the costs associated with certain operation is critical. The resulting expression that depicted the weighted average cost based on the information provided is depicted below.

$$Weighted\ Average\ Cost = \frac{\sum_i (RT\ Cost\ From\ Office)_i \cdot (Employees\ In\ Office)_i}{\sum_i (Employees\ In\ Office)_i}$$

Based on the previous relationships, this information may be utilized to find the total daily cost of walking. This expression accounts for the demand of each workstation that would occur throughout the duration of a typical work day. The resulting relationship that describes this is depicted in the equations shown below.

$$Demand \cdot (Weighted\ Average\ Cost) = Workstation\ Walking\ Cost\ Per\ Day$$

$$Total\ Daily\ Walking\ Cost = \sum_i (Workstation\ Walking\ Cost\ Per\ Day)_i$$

Using the relationships shown above, initial state data relating to distance, time, and cost was found in conjunction with the data collection activities. Where possible, the relevant dimensions were measured with a high level of precision to ensure that the findings appropriately depicted the system operations. The figure below shows a consolidated representation of these calculations.

	Office 1	Office 2	Office 3	Office 4	Office 5	Office 6	Office 7	Average	Weighted Average
Distance to 3D Printers	64.21'	75.04'	78'	78.16'	75.71'	63.42'	60.16'	70.67'	71.82'
Distance to Storage Room 1	56.21'	67.04'	70'	70.16'	67.71'	55.42'	52.16'	62.67'	63.82'
Distance to Storage Room 2	5.5'	2'	11.72'	15.88'	17.59'	17.72'	22.02'	13.2'	9.94'
Distance to 2D Printer	3.35'	16.57'	23.24'	25.3'	24.52'	18.11'	15.51'	18.09'	16.97'
Distance to Machine Room	57.71'	68.54'	71.5'	71.66'	69.21'	56.92'	53.66'	64.17'	65.32'
Time Roundtrip to 3D Printers	28.99 s	33.88 s	35.21 s	35.29 s	34.18 s	28.63 s	27.16 s	31.91 s	32.43 s
Time Roundtrip to Storage Room 1	25.38 s	30.27 s	31.6 s	31.67 s	30.57 s	25.02 s	23.55 s	28.29 s	28.81 s
Time Roundtrip to Storage Room 2	2.48 s	0.9 s	5.29 s	7.17 s	7.94 s	8 s	9.94 s	5.96 s	4.49 s
Time Roundtrip to 2D Printer	1.51 s	7.48 s	10.49 s	11.42 s	11.07 s	8.18 s	7 s	8.16 s	7.66 s
Time Roundtrip to Machine Room	26.05 s	30.94 s	32.28 s	32.35 s	31.25 s	25.7 s	24.23 s	28.97 s	29.49 s
Cost Roundtrip for 3D Printers	\$0.10	\$0.11	\$0.12	\$0.12	\$0.11	\$0.10	\$0.09	\$0.11	\$0.11
Cost Roundtrip for Storage Room 1	\$0.08	\$0.10	\$0.11	\$0.11	\$0.10	\$0.08	\$0.08	\$0.09	\$0.10
Cost Roundtrip for Storage Room 2	\$0.01	\$0.00	\$0.02	\$0.02	\$0.03	\$0.03	\$0.03	\$0.02	\$0.01
Cost Roundtrip for 2D Printer	\$0.01	\$0.02	\$0.03	\$0.04	\$0.04	\$0.03	\$0.02	\$0.03	\$0.03
Cost Roundtrip for Machine Room	\$0.09	\$0.10	\$0.11	\$0.11	\$0.10	\$0.09	\$0.08	\$0.10	\$0.10
Total Daily Walking Cost (Demand-based, in \$): 6.245074086									

Figure 3. Current State Calculated Data.

When handling complex systems, it is often beneficial to express various metrics in terms of weighted averages rather than depicted all of the individual values that comprise the average. As always, this is only suitable for application in circumstances where a weighted average would apply. In this case, visualizing the location of a weighted average on a diagram helps to arrive at alternative layouts which would be designed to minimize the non-value-added activities associated with movement. In many cases, a weighted average helps to visualize the general locations of workers within a system relative to the locations of the workstations. This is best expressed using a centroid. In definition [18], the centroid is a specific point which shows the geometric center of an object and is expressed as $(\bar{X}, \bar{Y})$. For the study at hand, the three centroids of interest are the weighted office opening centroid, the office opening centroid, and the Pod A centroid. The

formula for calculating the centroid is depicted below. The weight, “w,” is utilized to denote the weight associated with a particular component of the calculation.

$$C = (\bar{X}, \bar{Y})$$

$$\bar{X} = \frac{\sum_i x_i \cdot w_i}{\sum w}$$

$$\bar{Y} = \frac{\sum_i y_i \cdot w_i}{\sum w}$$

Through the effective application of the formulas shown above, these three centroids may be plotted on the initial state diagram in order to guide the analysis steps of the study. Naturally, these centroids are subject to change depending on the arrangement of the resources and equipment that are proposed for any given layout alternative. The plot containing a depiction of these centroids are shown in the diagram below.

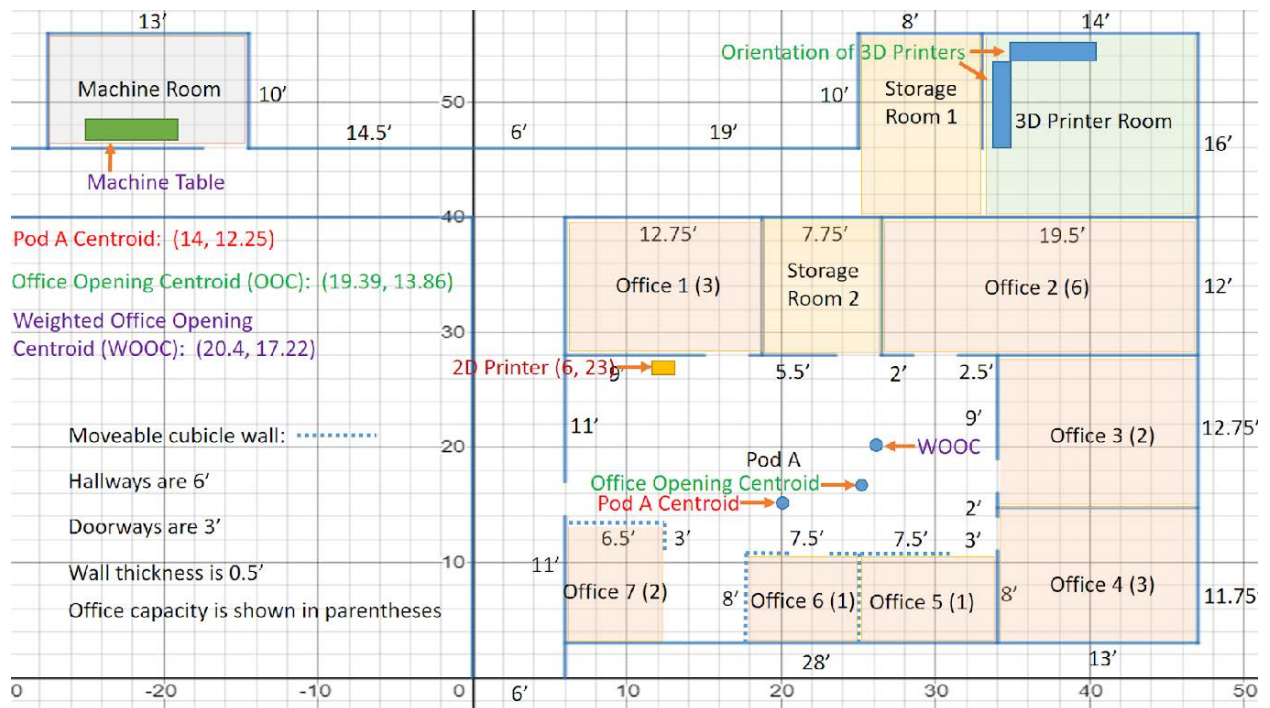


Figure 4. Initial State Centroids.

In reference to the diagram shown above, the Pod A centroid simply represents the center of the Pod A room. This centroid is useful in order to visually represent the center of the communal area so that alternative layouts may be structured around this. The Office Opening Centroid (OOC) represents the centroid of the office openings that are defined in the initial state of the system. Each office holds the same weight despite the number of employees which are housed in the office. Overall, this serves as a useful reference point that describes the current state of the system while also guiding future analysis activities. The Weighted Office Opening Centroid (WOOC) shares some elements of commonality with the office opening centroid, except the weights place on each office are directly related to the number of individuals housed within a respective office. Thus, among all three of the centroids proposed here, this centroid provides the most vivid and useful description of the current state of the system. An optimal solution (when ignoring qualitative concerns) would be to place all of the workstations at the location of the weighted office opening centroid. However, this change cannot be performed without infringing upon some of the highly important qualitative considerations that are valued by the governing stakeholders associated with the study.

When reviewing relevant scientific journals and literature, it became apparent that most layout studies carry a particular set of assumptions out of necessity. Likewise, some of these assumptions will also be relevant to the system at hand. The magnitude of the recommended changes should be reviewed at the conclusion of the study in conjunction with the ignored variables that were not included under the scope of the study at hand. Research has indicated that higher degrees of change are associated with stronger unexpected outcomes reaped from ignored variables along with a misalignment between the expectations and reality of the benefits associated with the implementation of the recommendations. Overall, this study seeks to describe the system

in a high level of detail in order to minimize the occurrences of ignored variables. To accomplish this purpose, an extensive data collection phase was implemented for the purpose of analysis. This phase included employee interviews, stakeholder preference elicitation using a variety of preference elicitation techniques, measurements, observations, and data tracking and recording over time. These insights combined with the informative results of the literature review resulted in a set of identified assumptions for the study at hand. Prior to enacting change, it is important to review the assumptions and constraints to ensure the accuracy of the results. As such, a set of constraints and associated assumptions are provided below.

Universal Constraints:

- Structural and fixed walls of the building cannot be moved or relocated in any fashion. In most cases, these walls are important for the structure of the building. Also, the incredible expense of moving such a structure would negate the positive impacts of any layout change.
- The UAH SMAP Center will not move to a different location to obtain a better layout for the outreach program. This is a reasonable assumption because of the close proximity between the working space of the outreach center and the associated university.
- Fire code regulations will be followed. In turn, this will shape the layout out of concern for the employees and the workplace.
- Placing the 2D or 3D printers in a room that does not also contain an employee would properly satisfy the air toxicity concern stated by OSHA. This constraint prevents an alternative layout from placing employees immediately next to a workstation due to the immense quantitative benefits associated with such a change.
- The types of machines which are present in the system will not change. This constraint prevents the type of machines from changing, but the number of machines is allowed to

fluctuate. Thus, the specific quantity of machines is not associated with the claim made in this constraint.

Variables that will be Ignored:

- Design changes that fulfill the previous set of universal constraints are assumed to also fulfill the OSHA regulations imposed upon additive manufacturing technologies in the workplace. It is unlikely that a proposed configuration will violate these regulations due to the strategically crafted qualitative factors of analysis which will be utilized to assess the suitability of each layout alternative.
- The power and routing requirements associated with any configuration are assumed to be manageable. This is directly due to the wealth of available power supplies and sources that are available within the system at hand.
- The HVAC circulation is assumed to be manageable and suitable for the configurations proposed. The resources presently residing in the system indicate that this is a reasonable assumption.
- Design changes for various alternative layouts are assumed to comply with fire code regulations which are issued by the city and university. Overall, interferences with key access areas for the purpose of fire code regulations is unlikely to arise with the alternative layouts being considered.
- The weight redistribution associated with an alternative layout and the resulting impact on the subtle deformations in the building and structural components will be ignored. Thus, the structural integrity of the building due to the shifting of people, machines, and office equipment will not be considered under the scope of analysis.

Factors Which are Subject to Change:

- The location of various workstations and offices may be changed along with the number of employees housed in each office. The movable cubicle walls may also be rearranged when proposing alternative layouts.
- The layout of the furniture in the system is subject to change in accordance with the needs of the various layout designs. Rearranging this furniture is not expected to yield any negative or unforeseen consequences.
- Rooms may be repurposed to suit various needs. Despite the initial state room assignments, a particular layout is allowed to redefine the purpose of each room in order to generate a favorable solution.
- A layout may enable greater amounts of expansion to hold more employees or resources in excess of what is allowed in the initial state of the system. This adds merit to an alternative layout due to enhanced capabilities.

Quantified Comparable Data:

- One important metric is the distance traveled from various offices to other areas of access. For this system, the data collection phase showed that walking from one workstation to another was rare. Thus, the only types of trips which were highly common were conducted directly between an office and a work station. The cost associated with this is expressed in the form of a total daily walking cost.
- Bonus features such as added storage space or increased space to house extra employees will be observed and noted, but the governing stakeholders would prefer to place the

strongest emphasis on reducing the total daily walking cost as a primary focal point for quantitative analysis.

- Other derived data will be utilized to characterize the various alternative layouts which are introduced throughout the study. This derived data is found from manipulation of the raw data retained from the data collection phase.

Qualitative (along with data that cannot be manifested accurately in a quantitative fashion)

Comparable Data:

- *Cost of Implementation.* This represents the cost associated with the labor and resources that would be necessary to actually achieve a particular layout alternative. Rearranging a layout is not exempt from costs, so this focal point is assessed from a qualitative perspective.
- *Mechanical Flexibility.* This assesses how difficult making adjustments to a configuration would be after it is fully implemented. Overall, it is desired to obtain a high level of flexibility in order to adapt to potentially changing circumstances.
- *Transparency of the Space.* This relates to the ease by which the operations of the office space may be observed by an individual. There are advantages and drawbacks to having a transparent space, but having a real-time update on the present condition of the system is often beneficial. The Transparency Paradox (examined in the literature review) plays a key role here even though it is counterintuitive.
- *Worker Privacy.* Worker privacy is an important qualitative factor because it relates to workplace motivation and human needs [19] [20] [21]. Detrimental impacts on worker privacy negatively correlate with impacts on productivity. As such, it is desired to obtain

a solution that does not compromise worker privacy to a significant extent to prevent losses in other areas of the work culture.

- *Overall Change from the Current State.* This is a qualitative estimation of how significantly a particular alternative layout would incur change in respects to the initial state of the system. This is an important focal point because higher levels of change often yield stronger unintended consequences arising from variables that are either ignored in the study or are not contained within the scope of the study. As always, it is important to advocate for positive change. However, this qualitative consideration seeks to hold an awareness of the likelihood of unforeseen outcomes.

4. Identify Feasible Alternatives:

One of the centerpieces of this study is an extensive literature review in the areas of work design theory, workplace motivation, and office layouts. Various scholarly sources were consulted including scientific journals, publications, and physical books describing research findings. The scope of this research ranged from the early 1960s to the modern era of office design trends. This historical perspective was pursued in order to provide a wide range of insight related to the subjects under consideration. As such, the applicable designs for office layouts that have been identified may be expressed as three broad domains.

1. Open Office Layouts
2. Systems Furniture Layouts (The Cubicle)
3. Combined Office Layouts (Based on a Data-Driven Design Integrated with Lean Principles)

Observing the evolution of each of these domains of office spaces over time has been insightful and meaningful for the development of this study. This research was important to provide a discussion of both the qualitative and the associated quantitative implications of various types of layouts. However, research relating strictly to office layouts from a quantitative perspective has become scarce in the modern era. Many of the quantitative approaches to assessing office spaces were originally pioneered in the 20th Century, specifically following the beginning of the 1960s. Turban and Bernstein wrote that “to our knowledge, we are the first to empirically study the direct behavior impact of open office space” in their 2018 examination of two Fortune 500 companies who initiated the transition from a Systems Furniture layout to an Open Office layout. Given this, the study will proceed to examine alternative layouts that are directly inspired by the indications and findings of the literature review.

The Open Office Layout:



Image Link Citation

Typical Bank Office



Image Link Citation

Frank Lloyd Wright's Johnson Wax Headquarters 1930



Image Link Citation

Office Scene from 1960 film “The Apartment”



Image Link Citation

“Modern” Open Office design from a construction firm

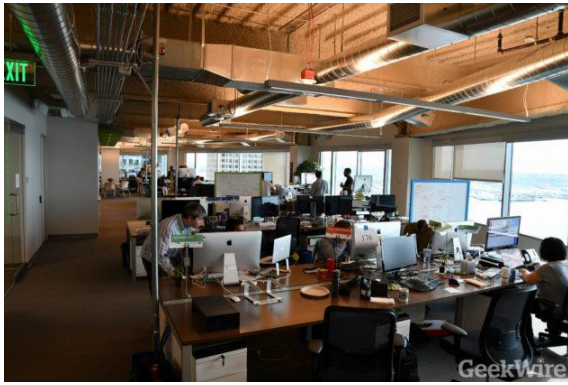


Figure 5. Depictions of Open Offices [5] [6] [7] [8] [9].

Fascinatingly, the open office layout is considered to be the oldest of the three layouts entailed under the domain of this study. One instrumental historical moment for the open office was the Burolandschaft (also known as the Office Landscape Movement) which took place in Germany in the 1950s [10] [22]. This movement proposed a rejection of Taylorism, which was pioneered by Frederick Winslow Taylor in his book “Principles of Scientific Management.” Instead, employees desired for office spaces to place a central focus in the strategic design process on the flow of information in the workplace. In turn, companies adapted and these changes still

hold influence on office layouts to this day. From a broad perspective, there are a total of three key aspects that characterize a typical open office layout.

High Utilization of Space:

The open office directly targets efficiency in several regards. This layout seeks to cluster personal spaces together in rows of desks, each assigned to one person, in order to maximize the ratio of communal space to personal space. Instead of having an entire room for an office, each person simply has one small area that may be mildly considered as personal. An open office philosophy would regard a traditional office room as having a high level of under-utilized space. In the traditional manifestation of an open office layout, the workers do not have very much spacing separating them. While the utilization of such an office space remains high (to a point that approaches optimality), there are qualitative concerns associated with such a layout that may deter some companies from adopting such a layout.

“You have a lot of problems with departments when you take away all their private offices – and we sure did; we eliminated about 127 private offices. But in return we gave them all nice, new, air-conditioned, wide office space, new furniture, and new surroundings. You take something, and you give something.” --- Albert N. McGrath

“As clerical tasks are replaced, machines and the positions of machines become a significant factor in space planning. The designer’s solution must take into account the change from the individual desk to the continuous row-type workstation unit which houses both the machines necessary to accomplish the work and the clerical work position.” --- Donald D. Powell

Universality of the Workspace:

Within the domain and space of an open office layout, a worker does not have a private office. Moreover, the worker does not have a specified desk or area that may be considered as personal. All spaces are communal to varying capacities and this impacts the work culture. The emergent outcome of this structure is threefold. First, team formation becomes more organic due to the natural grouping and clustering of labor, thereby increasing interpersonal interfacing. Second, the workplace remains in an optically ordered fashion due to the fact that the natural clutter of personal spaces is minimized. Third, the expected utilization and efficiency of the operational function of the workspace increases particularly when not all employees are actively present simultaneously. Ultimately, these three accomplishments characterize the nature of the open office layout.

“The office personality will take on a totally new character, with a departure from the traditional proprietary idea of ‘this is my desk, my typewriter, my chair.’ I foresee the new workstation designed around the principle of great flexibility in moving people about, so that they have no set position. This is going to bring many problems, because I, like everyone else, make a home of my desk - which I think is totally wrong. There will be greater, not less, people-to-people contact in the office of the future.” --- Donald D. Powell

“There are very few provisions for personal items. We did extensive research on this point – the girls had purses, and their winter shoes, and their bowling shoes, and all sorts of things in their desks. We have provided a secretarial position locker in the basement of this building where the girl can have her coat and her bowling shoes and her bowling ball and her make-up and all the rest, so she does not have personal items...(at the individual workstation).” --- Donald D. Powell

Shared Access to Machines and Communal Spaces:

The open office layout places a strong focus on communal spaces. As widely noted in literature, a phenomenon known as The Tragedy of the Commons often arises in communal spaces such as those proposed in an open office plan [23] [24] [25]. In summary, when individuals perceive the maintenance of a space as having communal responsibility rather than personal responsibility, the conditions associated with the communal aspects of the space tend to erode over time unless someone is appointed to take responsibility of the maintenance of such as space. Qualitatively speaking, this is an observable drawback in the enactment of an open office space for most companies. Shown below is a depiction of the open office concept as applied to the system of interest.

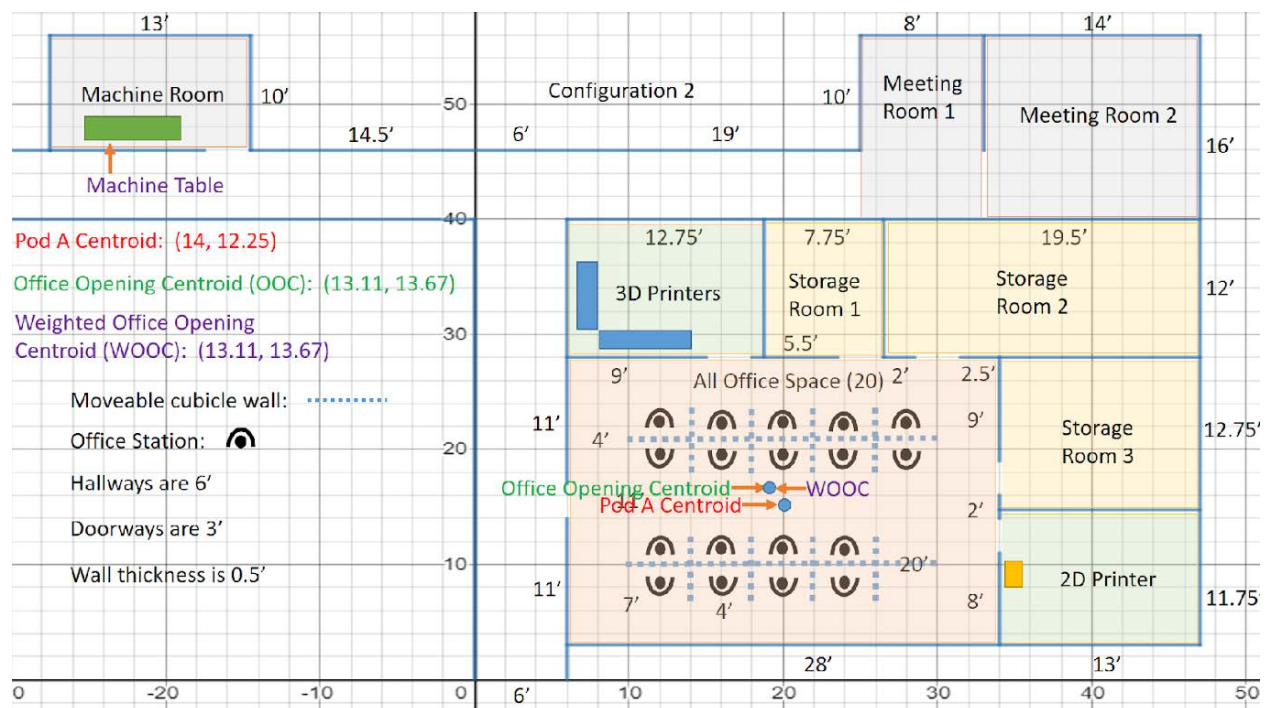


Figure 6. Open Office Manifestation of the System.

As shown in the figure depicted above, the areas of access (workstations) share a much closer proximity with the various centroids that were proposed for the purpose of analysis. This space would succeed the most in a team environment where minimal personal space is an acceptable outcome. In total, the changes to the system from the initial state that were required in order to construct the open office concept are detailed below.

- Move the office items located in the various offices into the area for Pod A
- Orient the cubicle walls in the arrangement depicted in the figure above
- The 2D printer was moved into the space which was previously Office 4
- All of the 3D printers were moved into the space which was previously Office 1
- Items contained in Pod A in the initial state were moved to the original Storage Room 1 and the 3D Printer Room
- Offices 2, 3, and 4 in the initial state now hold storage items collected from the various storage rooms depicted in the initial state diagram

At this point, the layout under consideration will be analyzed under the qualitative measures proposed and emphasized by the governing stakeholders of the system at hand. This involves examining the suggested layout under the context of each qualitative consideration. Ultimately, the considerations contained under the domain of this qualitative analysis cannot be quantified or rigidly ranked. As such, judgment will be applied in accordance with the perception of the system under the circumstances proposed in a given layout alternative. The overall discussion and commentary shown below describes the qualitative implications of this particular layout alternative.

Qualitative Measures:

In total, this layout will be assessed on the basis of five qualitative considerations in order to determine the suitability of the layout in regards to the stakeholder preferences which were captured. These considerations consist of cost of implementation, mechanical flexibility, transparency of the space, worker privacy, and overall change from the current state. The results from assessing the proposed layout alternative are depicted in the figure shown in the space provided below.

Layout Qualitative Factors	Open Office Layout
Cost of Implementation	Good
Mechanical Flexibility	Excellent
Transparency of the Space	Excellent
Worker Privacy	Inferior
Overall Change from the Current State	Most

Figure 7. Qualitative Results for the Open Office Layout.

Given this establishment, the proposed layout will be analyzed from a quantitative perspective. While there are a number of quantitative approaches that may be applied in order to assess the suitability of a particular layout, the key metrics that were distinguished as critical by the governing stakeholders will be examined in order to form a quantitative foundation of analysis in regards to the associated alternative layout. The results of this quantitative analysis are depicted in the following section.

Quantitative Measures:

To begin, the previous quantitative techniques discussed were evaluated under the context of this particular layout. These assessments are a measure of distance, time, and cost related to non-value-added activities. Ultimately, these calculations are summed to yield a value which represents the total daily cost of walking. The results were then compiled and presented in the figure shown below.

Open Office:	Average Distance	Average RT Time	Average RT Cost (\$)	Daily Cost of Walking Time (\$)
3D Printers (41 accesses/day)	18.16'	8.2 s	0.027336135	1.120781546
Storage (20 accesses/day)	17.56'	7.93 s	0.026432546	0.528650839
2D Printer (16 accesses/day)	15.61'	7.05 s	0.023488207	0.375811318
Machines (3 accesses/day)	53.79'	24.28 s	0.080949326	0.242847977
			Total Daily Walking Cost (\$): 2.26809168	

Figure 8. Quantitative Results for the Open Office Layout.

At this point, it is desired to analyze the layout on the basis of an effectiveness calculation. This calculation will be assessed as a multiplication of the distance from each office to each workstation and the number of times that the trip is expected to be travelled in any given work

day. The summation of these effectiveness values would yield the expected distance travelled by all employees (between offices and workstations) in a regular work day of operations. In this system, the data collection phased showed that travelling from one workstation to another was very rare. Thus, trips were only conducted between an office and a workstation. Due to this phenomenon, the traditional representation of the effectiveness calculations will need to be altered to account for this fact. As such, the beginning distances chart is depicted in the figure shown in the space provided below.

Open Office Layout	3D Printers	Storage	2D Printer	Machines
Weighted Office Opening Centroid (WOOC)	18.16'	17.56'	15.61'	53.79'

Figure 9. Average Distances from a Desk to a Workstation.

Given the distances chart shown above, the from-to chart for this particular layout will be obtained. In order to determine the overall effectiveness of the layout from a quantitative perspective, the individual values in the cells of the table which depicts the average distances will be multiplied by the values in the cells of the from-to chart. The results of these calculations will be summed in order to yield a total effectiveness for the layout under consideration. The from-to chart is depicted in the figure shown below.

From ↓ To →	3D Printers	Storage	2D Printer	Machines
Weighted Office Opening Centroid (WOOC)	41	20	16	3

Figure 10. From-To Chart (in Number of Accesses per Day) for the System.

Given the from-to chart shown in the figure above, the total effectiveness of the layout will be obtained. Typically, the workstations which have the highest number of accesses per day or the longest distance travelled with contribute the most to the final value for effectiveness due to the method by which effectiveness is calculation. The figure below depicts the final calculations pertaining to effectiveness for this particular layout.

From ↓ To →	3D Printers	Storage	2D Printer	Machines	Total
Weighted Office Opening Centroid (WOOC)	$18.16 \cdot 41$	$17.56 \cdot 20$	$15.61 \cdot 16$	$53.79 \cdot 3$	1506.89
				Total Effectiveness:	
				1506.89	

Figure 11. Effectiveness Assessment of the Open Office Layout.

At this stage of the quantitative analysis pertaining to the layout at hand, simulation will be applied to assess the layout. The primary focal point of the application of simulation is to gain

an understanding of the variability surrounding the travel times within the layout overall. To this point, travel time has been regarded as fixed and deterministic. Realistically, this is not the case. Employees have subtly varying walking speeds that should be accounted for. Due to the low volume in demand for the various work stations, the length of the queues within the system are typically zero (or a length of one in a rare case). Theoretical calculations under the domain of queueing theory verify this observable phenomenon. Simio was selected for application on this layout. In the simulation model, walking speed was modeled (involving $D = R \cdot T$) as a normally distributed (in the form of $X \sim N(\mu, \sigma^2)$) random variable in the generalized form of $R \sim N(4.43, 0.443^2)$. The figure below depicts the associated Simio simulation model for the layout under consideration.

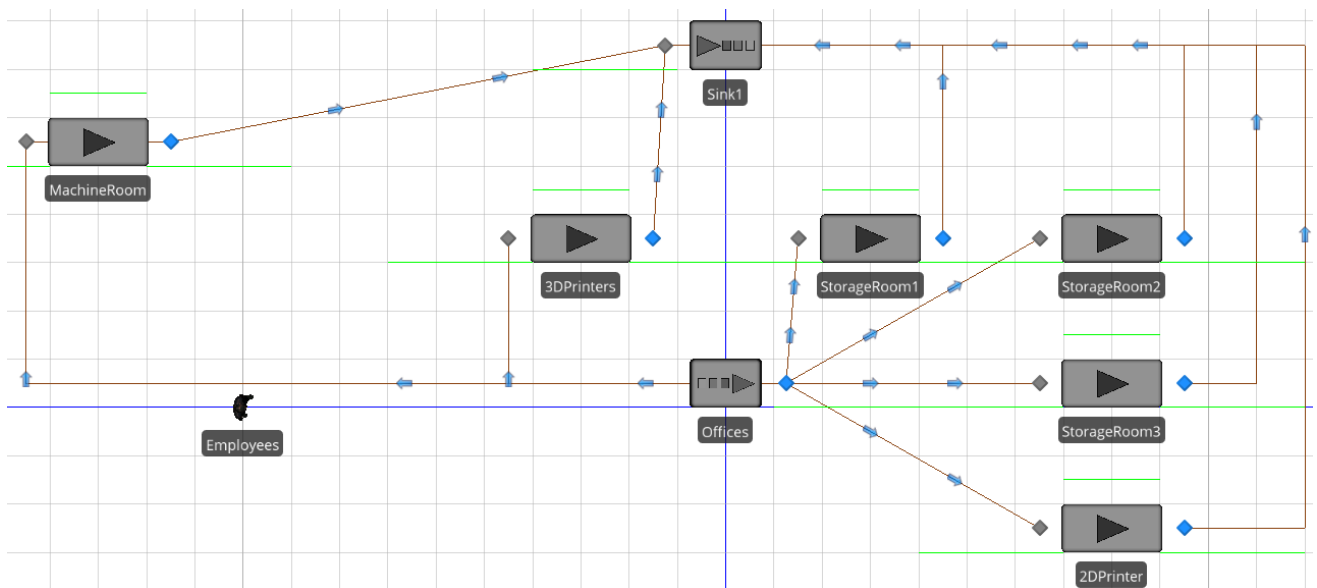


Figure 12. Simio Simulation Model for the Open Office Layout.

Based on this model, an experiment was constructed using a total of 100 replications of the system, where each replication analyzed 2 hours of operational data. The mean time for a roundtrip (weighted based on the demand for each workstation) according to the simulation model was

8.551655 seconds with a standard deviation of 0.335808 seconds. Hand calculations show that the expected value (underlying mean) of this time is about 8.5053893905 seconds. The figure below depicts the histogram and the associated box plot for the response statistic defined to be the average roundtrip time.

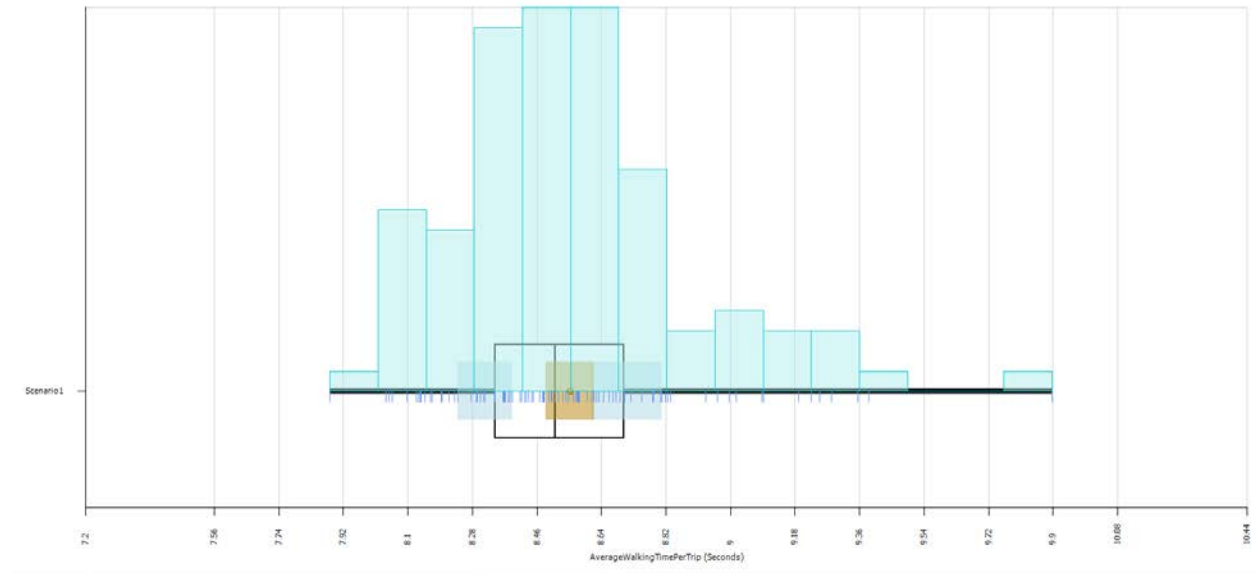


Figure 13. Histogram of the Average Walking Time Per Trip Statistic (n = 100).

At this point, it is desired to assess the validity of the simulation model by checking the consistency between the sampled data and the hand calculations of the theoretical mean value. This may be analyzed through the application of a hypothesis test. A typical hypothesis test for large samples will be applied to make this determination. In this case, the value for σ is not known, so the standard deviation, s , will be substituted. The resulting calculations of the hypothesis test are provided below.

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0$$

$$z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \quad (s \text{ may substitute for } \sigma)$$

$$H_0: \mu = 8.5053893905$$

$$H_1: \mu \neq 8.5053893905$$

$$z = \frac{8.551655 - 8.5053893905}{\frac{0.335808}{\sqrt{100}}} = 1.4223$$

$$P - \text{value} \approx 0.1549$$

The results of the hypothesis test above indicate that the null hypothesis cannot be rejected when using a level of significance such as $\alpha = 0.05$. Thus, it is reasonable to conclude that the two mean values are equivalent within 95% confidence. Therefore, the simulation model constructed provides an accurate representation of the alternative layout proposed on the basis of this quantitative information. Anomalies are unlikely to arise due to random variation over the course of 100 replications lasting 2 hours each.

Next, this layout will be analyzed to assess the proportion of task operation time that should be generally expected under this layout. This will be calculated by dividing the operation time by the total time. In this case, the total time is defined as the sum of the operation time and the roundtrip travel time to reach the location of a workstation. The results of these calculations are provided in the figure shown below.

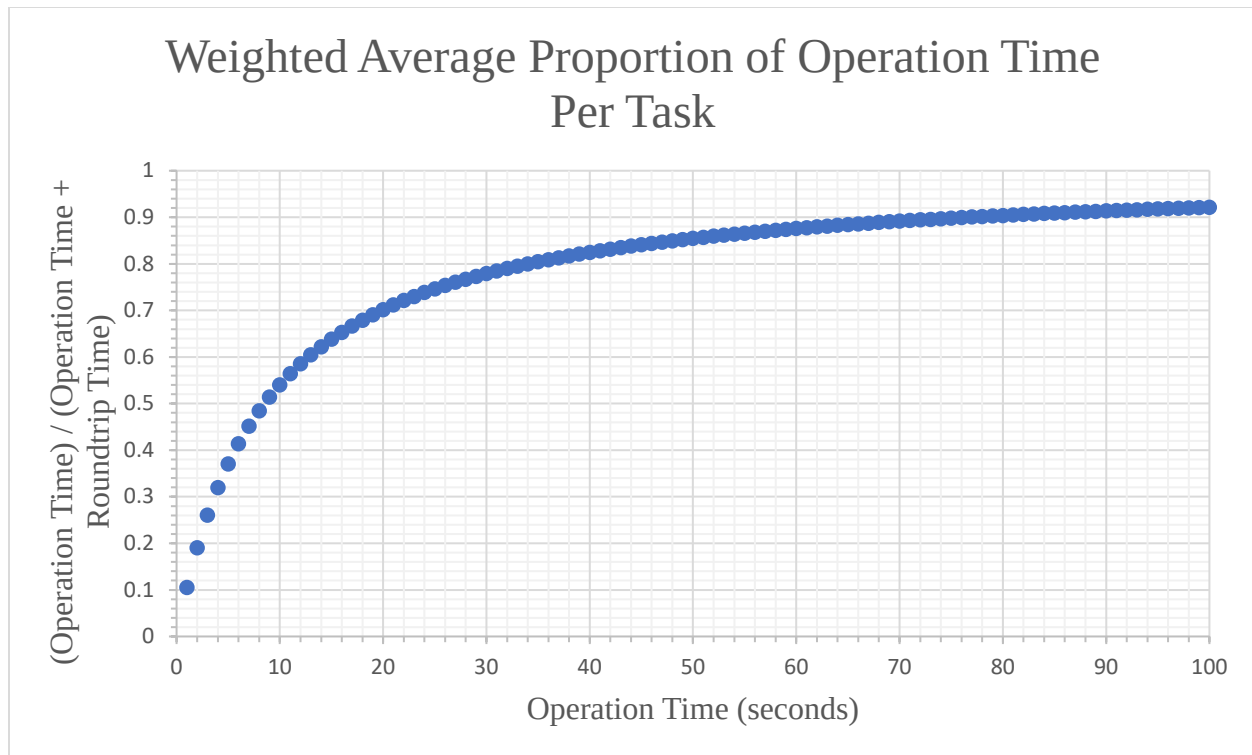


Figure 14. Weighted Average Proportion of Operation Time per Task.

This concludes the quantitative analysis procedures which will be performed on this particular layout. The insights gained from the manipulation of raw data, the Simio simulation model, and the expected proportion of operation time will be used as the foundation of analysis for the layout. Furthermore, the various alternative layouts will ultimately be compared in these regards in order to determine the layout which would best address the governing stakeholder needs as previously expressed.

The Systems Furniture Layout (The Cubicle):



Image Link Citation

Robert Probst's Action Office 2 circa 1970

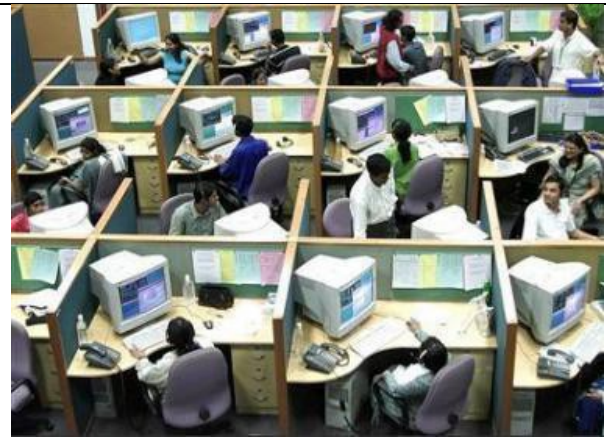


Image Link Citation

1990 low wall Cubicle Office layout



Image Link Citation

Common high wall Cubicle implementation

1980 common parlance "Cube Farm"



Image Link Citation

Scene from the 1999 movie "Office Space"

Showing the common implementation of the mid wall design

Figure 15. Systems Furniture (The Cubicle) Layout Examples [11].

The historical gravitation towards open office plans and the subsequent failings led to a reactionary motivation to implement the systems furniture layout. The early adoption of the systems furniture layout (also known as the cubicle) emerged in the 1960s timeframe. Robert Probst led the development of this layout when he designed layouts for the Herman Miller firm called the Action Office and Action Office 2 layouts. In the development of these layouts, Probst

held very specific goals in mind. Ultimately, he desired to “give knowledge workers a more flexible, fluid environment” through enhancing interpersonal interactions, minimizing distractions, and increasing privacy in manners by which the open office failed to capture [12]. This was a key component of a much broader shift in the prevailing work culture of society that sought to grant workers more autonomy. This type of broad shift began as a reaction to the shortcomings of the working conditions experienced throughout the first industrial revolution in the early 1800s.

The innovative designs produced by Robert Probst yielded a layout which featured private space for individuals separated with partitions in a modular fashion. This widely became known as the systems furniture layout, or the cubicle. While the most common orientation of the partitions are intersections at a 90-degree angle, other designs involved a 120-degree orientation between the partitions. He often claimed that “the cubicle-izing of people in modern corporations is monolithic insanity.” A 1960s change in the governing United States tax code also offered a favorable hand to the adoption of the systems furniture layout. This change allowed for the depreciation of office equipment over the course of 7 years rather than the previous 39.5 years for permanent structures. Thus, this change encouraged business owners and managers to adopt the systems furniture layout in order to take advantage of the depreciation benefits offered through the updated United States tax code laws. From a broad perspective, the systems furniture layout holds three beneficial characteristics.

Individually Assigned Workstations:

In contrast to the open office layout, the systems furniture layout features individually assigned offices. However, these offices are quite small and minimalistic. In some cases, the systems furniture layout may offer cubicles as a primary means of allowing workers to focus while

completing tasks. Alternatively, the cubicles may simply function as an area to store personal items while workers perform tasks in other locations. Under either circumstance, workers hold higher levels of private space in the systems furniture layout.

“It is our conviction that the type of furniture and equipment provided to an individual should be exactly what he requires to handle the task at hand most effectively – no more no less. This, then, determines the amount of space required to house the workstation.” --- Walter Waddel

Vertically-Integrated Work Surfaces (Partitions):

Partitions may come in the form of an informal (or moveable) wall or something more natural such as a bookshelf to separate two office spaces. These types of dividers are highly prevalent in systems furniture layouts around the globe. Products have been strategically developed and sold to the mass markets in order to address the needs of a systems furniture layouts. In any circumstance where numerous partitions separate individual workspaces, it is highly likely that this arrangement is inspired or derived from the systems furniture layout. While open offices may also contain these partitions, it is far more common for an open office plan to target functionality rather than privacy.

Prioritization of Workspace Over Utilization:

The systems furniture layout does not place a relentless focus on the utilization of space as the open office layout does. However, the systems furniture layout does strive to create a high utilization for private office space. Typically, this layout seeks to minimize the non-value-added space within a personal office. Unlike the open office, the systems furniture layout recognizes the importance of having a private office for workers. Shown in the figure below is a manifestation

of the principles and concepts of the systems furniture layout as applied to the system under consideration.

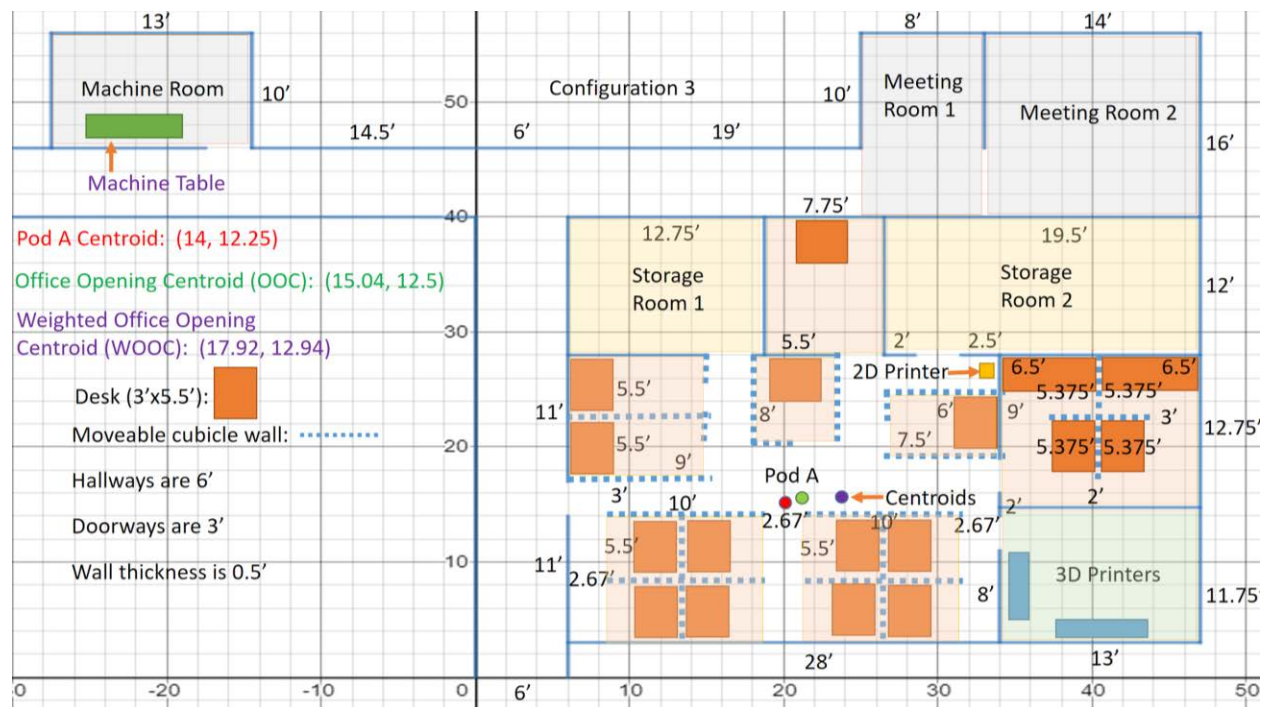


Figure 16. Systems Furniture Layout Concept for the System.

The design shown in the figure above strives to maintain the collaborative atmosphere that the open office offers while also enabling employees to have some amount of private office space. While the compatibility of this design with collaborative work is less than that of the open office, it is reasonable to conclude that both individualistic and collaborative work would be suitable for this type of layout. However, it is important to understand the resulting impact on morale due to the low amount of private office space. If employees are instructed to carry out tasks primarily from their own offices, then morale may drop due to the decrease in outside interaction. However, this layout would likely have positive impacts on morale if a healthy mix of tasks were conducted

outside of the defined private office space. The list of changes required to implement the proposed systems furniture layout are described below.

- Move the 3D printers from the 3D printer room to Office 4 (as defined in the initial state)
- Transition all storage items to be housed in Offices 1 and 3 (these rooms become the new storage rooms)
- Rearrange the moveable cubicle walls to fit the orientation shown in the figure shown above
- Move the items originally in Pod A to the initial state definitions for the 3D printer room and Storage Room 1 (become Meeting Rooms 1 and 2)
- Arrange the office equipment (desks, chairs, and other furniture) to align with the orientations depicted in the figure shown above

At this point, the layout under consideration will be analyzed under the qualitative measures proposed and emphasized by the governing stakeholders of the system at hand. This involves examining the suggested layout under the context of each qualitative consideration. Ultimately, the considerations contained under the domain of this qualitative analysis cannot be quantified or rigidly ranked. As such, judgment will be applied in accordance with the perception of the system under the circumstances proposed in a given layout alternative. The overall discussion and commentary shown below describes the qualitative implications of this particular layout alternative.

Qualitative Measures:

In total, this layout will be assessed on the basis of five qualitative considerations in order to determine the suitability of the layout in regards to the stakeholder preferences which were

captured. These considerations consist of cost of implementation, mechanical flexibility, transparency of the space, worker privacy, and overall change from the current state. The results from assessing the proposed layout alternative are depicted in the figure shown in the space provided below.

Layout Qualitative Factors	Systems Furniture Layout
Cost of Implementation	Poor
Mechanical Flexibility	Poor
Transparency of the Space	Poor
Worker Privacy	Excellent
Overall Change from the Current State	Moderate

Figure 17. Qualitative Results for the Systems Furniture Layout.

Given this establishment, the proposed layout will be analyzed from a quantitative perspective. While there are a number of quantitative approaches that may be applied in order to assess the suitability of a particular layout, the key metrics that were distinguished as critical by the governing stakeholders will be examined in order to form a quantitative foundation of analysis in regards to the associated alternative layout. The results of this quantitative analysis are depicted in the following section.

Quantitative Measures:

To begin, the previous quantitative techniques discussed were evaluated under the context of this particular layout. These assessments are a measure of distance, time, and cost related to non-value-added activities. Ultimately, these calculations are summed to yield a value which

represents the total daily cost of walking. The results were then compiled and presented in the figure shown below.

Systems Furniture:	Average Distance	Average RT Time	Average RT Cost (\$)	Daily Cost of Walking Time (\$)
3D Printers (41 accesses/day)	12.2'	5.51 s	0.018359669	0.0752746426
Storage (20 accesses/day)	17.49'	7.9 s	0.026323007	0.526460143
2D Printer (16 accesses/day)	18.47'	8.34 s	0.027802494	0.444839901
Machines (3 accesses/day)	59.36'	26.8 s	0.089333459	0.268000377
			Total Daily Walking Cost (\$): 1.992046847	

Figure 18. Quantitative Results for the Systems Furniture Layout.

At this point, it is desired to analyze the layout on the basis of an effectiveness calculation. This calculation will be assessed as a multiplication of the distance from each office to each workstation and the number of times that the trip is expected to be travelled in any given work day. The summation of these effectiveness values would yield the expected distance travelled by all employees (between offices and workstations) in a regular work day of operations. In this system, the data collection phased showed that travelling from one workstation to another was very rare. Thus, trips were only conducted between an office and a workstation. Due to this phenomenon, the traditional representation of the effectiveness calculations will need to be altered to account for this fact. As such, the beginning distances chart is depicted in the figure shown in the space provided below.

Systems Furniture Layout	3D Printers	Storage	2D Printer	Machines
Weighted Office Opening Centroid (WOOC)	12.2'	17.49'	18.47'	59.36'

Figure 19. Average Distances from a Desk to a Workstation.

Given the table shown in the figure above, the total effectiveness of the layout will be obtained. Typically, the workstations which have the highest number of accesses per day or the longest distance travelled with contribute the most to the final value for effectiveness due to the method by which effectiveness is calculation. The figure below depicts the final calculations pertaining to effectiveness for this particular layout.

To → From ↓	3D Printers	Storage	2D Printer	Machines	Total
Weighted Office Opening Centroid (WOOC)	12.2 · 41	17.49 · 20	18.47 · 16	59.36 · 3	1323.6
				Total Effectiveness: 1323.6	

Figure 20. Effectiveness Assessment of the Systems Furniture Layout.

At this stage of the quantitative analysis pertaining to the layout at hand, simulation will be applied to assess the layout. The primary focal point of the application of simulation is to gain an understanding of the variability surrounding the travel times within the layout overall. To this point, travel time has been regarded as fixed and deterministic. Realistically, this is not the case. Employees have subtly varying walking speeds that should be accounted for. Due to the low volume in demand for the various work stations, the length of the queues within the system are typically zero (or a length of one in a rare case). Theoretical calculations under the domain of queueing theory verify this observable phenomenon. Simio was selected for application on this layout. In the simulation model, walking speed was modeled (involving $D = R \cdot T$) as a normally distributed (in the form of $X \sim N(\mu, \sigma^2)$) random variable in the generalized form of $R \sim N(4.43, 0.443^2)$. The figure below depicts the associated Simio simulation model for the layout under consideration.

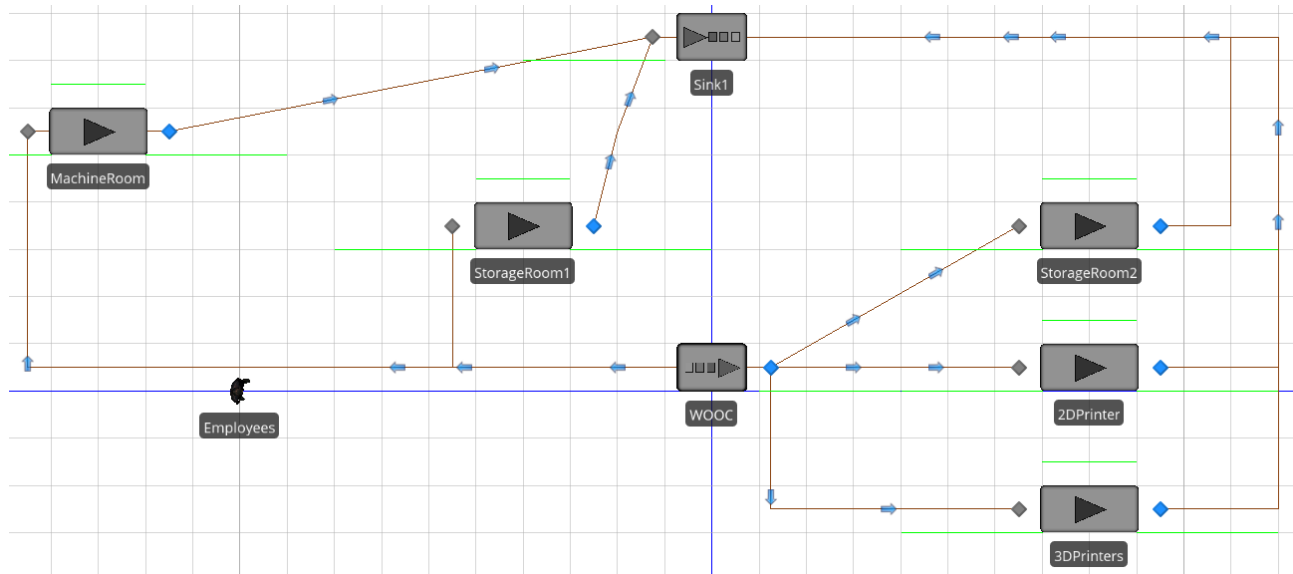


Figure 21. Simio Simulation Model for the Systems Furniture Layout.

Based on this model, an experiment was constructed using a total of 100 replications of the system, where each replication analyzed 2 hours of operational data. The mean time for a roundtrip (weighted based on the demand for each workstation) according to the simulation model was 7.49111 seconds with a standard deviation of 0.469955 seconds. Hand calculations show that the expected value (underlying mean) of this time is about 7.469525959 seconds. The figure below depicts the histogram and the associated box plot for the response statistic defined to be the average roundtrip time.

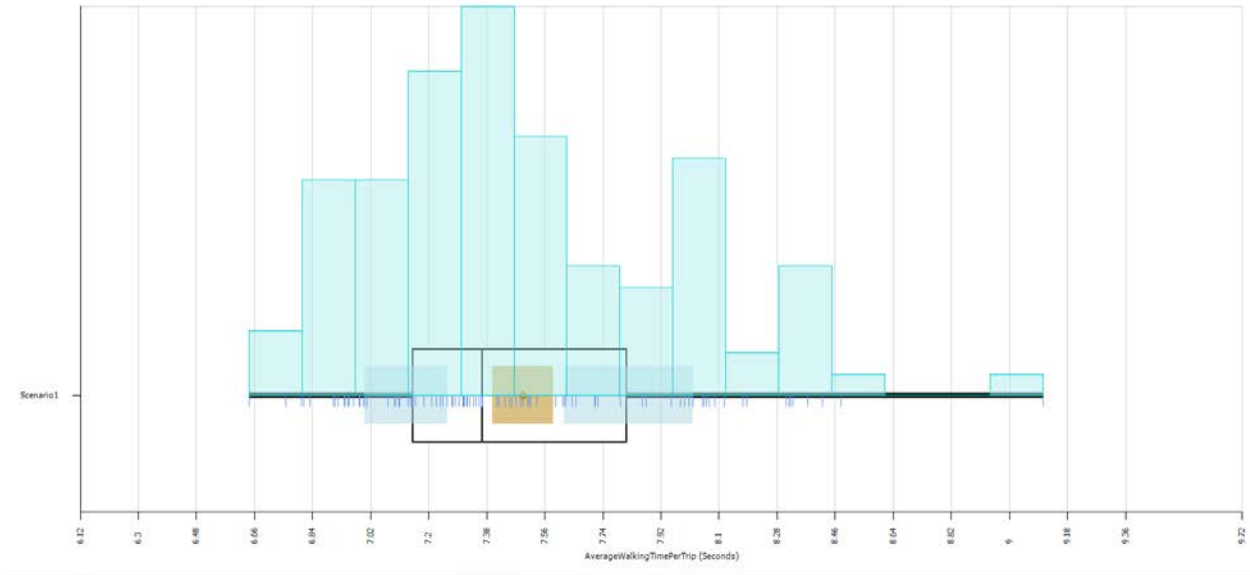


Figure 22. Histogram of the Average Walking Time Per Trip Statistic (n = 100).

At this point, it is desired to assess the validity of the simulation model by checking the consistency between the sampled data and the hand calculations of the theoretical mean value. This may be analyzed through the application of a hypothesis test. A typical hypothesis test for large samples will be applied to make this determination. In this case, the value for σ is not known, so the standard deviation, s , will be substituted. The resulting calculations of the hypothesis test are provided below.

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0$$

$$z = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}} \quad (s \text{ may substitute for } \sigma)$$

$$H_0: \mu = 7.469525959$$

$$H_1: \mu \neq 7.469525959$$

$$z = \frac{7.49111 - 7.469525959}{\frac{0.469955}{\sqrt{100}}} = 0.4593$$

$$P - value \approx 0.646$$

The results of the hypothesis test above indicate that the null hypothesis cannot be rejected when using a level of significance such as $\alpha = 0.05$. Thus, it is reasonable to conclude that the two mean values are equivalent within 95% confidence. Therefore, the simulation model constructed provides an accurate representation of the alternative layout proposed on the basis of this quantitative information. Anomalies are unlikely to arise due to random variation over the course of 100 replications lasting 2 hours each.

Next, this layout will be analyzed to assess the proportion of task operation time that should be generally expected under this layout. This will be calculated by dividing the operation time by the total time. In this case, the total time is defined as the sum of the operation time and the roundtrip travel time to reach the location of a workstation. The results of these calculations are provided in the figure shown below.

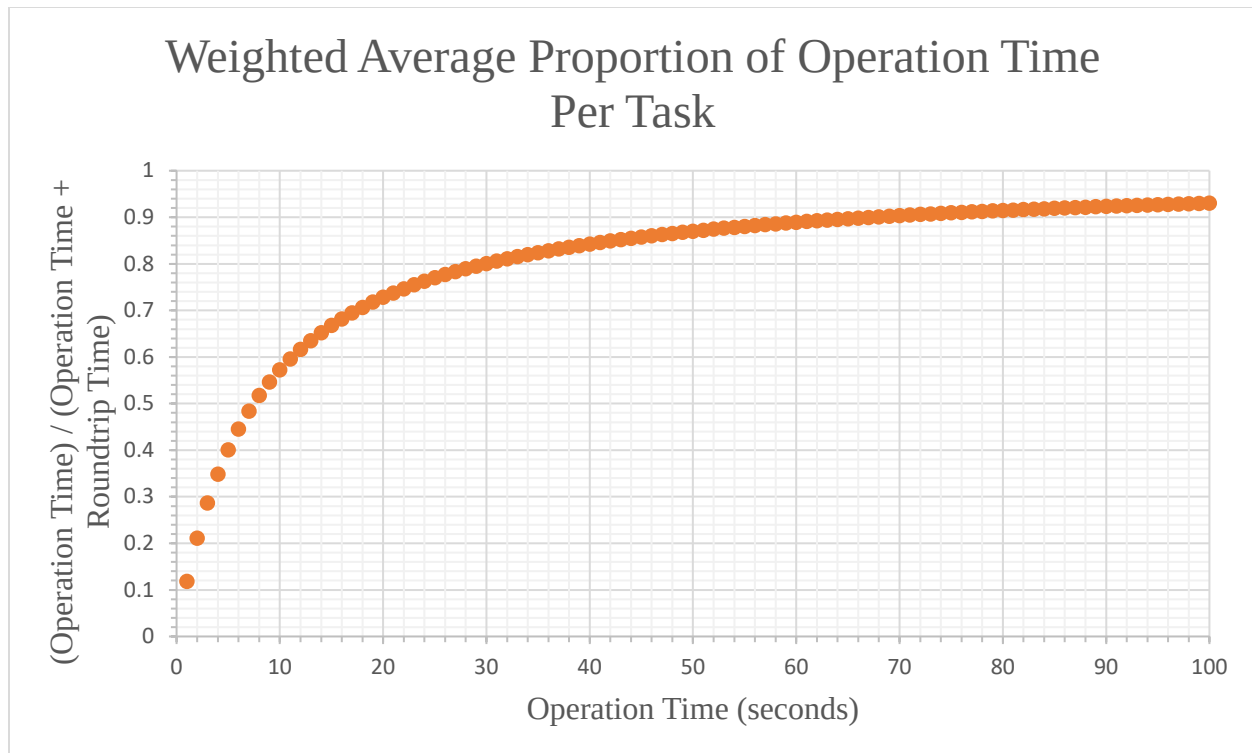


Figure 23. Weighted Average Proportion of Operation Time per Task.

This concludes the quantitative analysis procedures which will be performed on this particular layout. The insights gained from the manipulation of raw data, the Simio simulation model, and the expected proportion of operation time will be used as the foundation of analysis for the layout. Furthermore, the various alternative layouts will ultimately be compared in these regards in order to determine the layout which would best address the governing stakeholder needs as previously expressed.

The Combined Offices (Data-Driven Design with Lean Principles):



Figure 24. Examples of Combined Office Layouts [14].

In contrast to the previous two office layouts, the combined office layout does not have a universal or underlying theme in terms of primary objectives. Rather, the principles and objectives of the combined office are typically tailored to address situational circumstances (centered around the type of work performed in the office space) while also capturing the benefits offered by elements of the open office and systems furniture layouts. Due to this fluidity, the planning stage of the combined office layout requires the most amount of time among the three layouts proposed in this study. Due to the heavy burden imposed on the planning stage of a typical combined office layout, this layout is regarded as being the least used (among the three layouts proposed in this study) in modern society. The underlying inspiration for the combined office layout is strongly rooted in the Lean methodology and philosophy as pioneered by the Toyota Production System. More broadly speaking, this layout holds important crossovers and considerations with modern industrial work methods. For the specific context under consideration, the primary objectives which are defined for the combined office layout includes an allowance for flexibility, an accommodation for the type of work performed, a minimization of wasted motions and

movements, and a healthy amount of caution towards changes which are not directly informed by data-driven insight.

Eliminate Wasted Movements:

Under the context in consideration, the combined office places a unique focus on minimizing wasted movements associated with the operational activities of the system. This focus also falls in stride with a key motivation imbedded in the Lean methodology. This goal of the combined office is achieved through the strategic minimization of travel distances to workstations while also thoughtfully selecting the locations of workstations based on team needs. Furthermore, qualitative considerations influence the extent to which wasted movements may be reduced in the system of interest.

Tailor Spaces to the Work Performed:

The objective seeks to strike a balance between compatibility with team efforts and individualistic work. The layout seeks to satisfy both of these needs in order to ensure that the operations of the system do not experience any drawbacks on the basis of productivity or communication. The space is specifically intended to enable areas for regional collaboration when also allowing area for private office spaces. While the type of work performed is subject to change over time, the layout is intended to be flexible enough to accommodate these changes in work requirements.

Maximum Flexibility:

The type of flexibility that the combined office attempts to capture relates to flexibility in the type of work performed as well as the capability to rearrange office equipment when needed to suit differing purposes. While other layouts may be immune to changes in the type of work

performed, the combined office is fundamentally centered around capturing the capability. The malleability of a workspace is often pursued when changes in work are expected over time. For this reason, many workplaces would not be tempted to pursue a combined office layout if the work activities typically remain fixed and constant over time. Due to the everchanging nature of the work performed within this system, having an element of flexibility would be a beneficial feature of a layout. Shown in the figure below is a combined layout concept for the system under consideration.

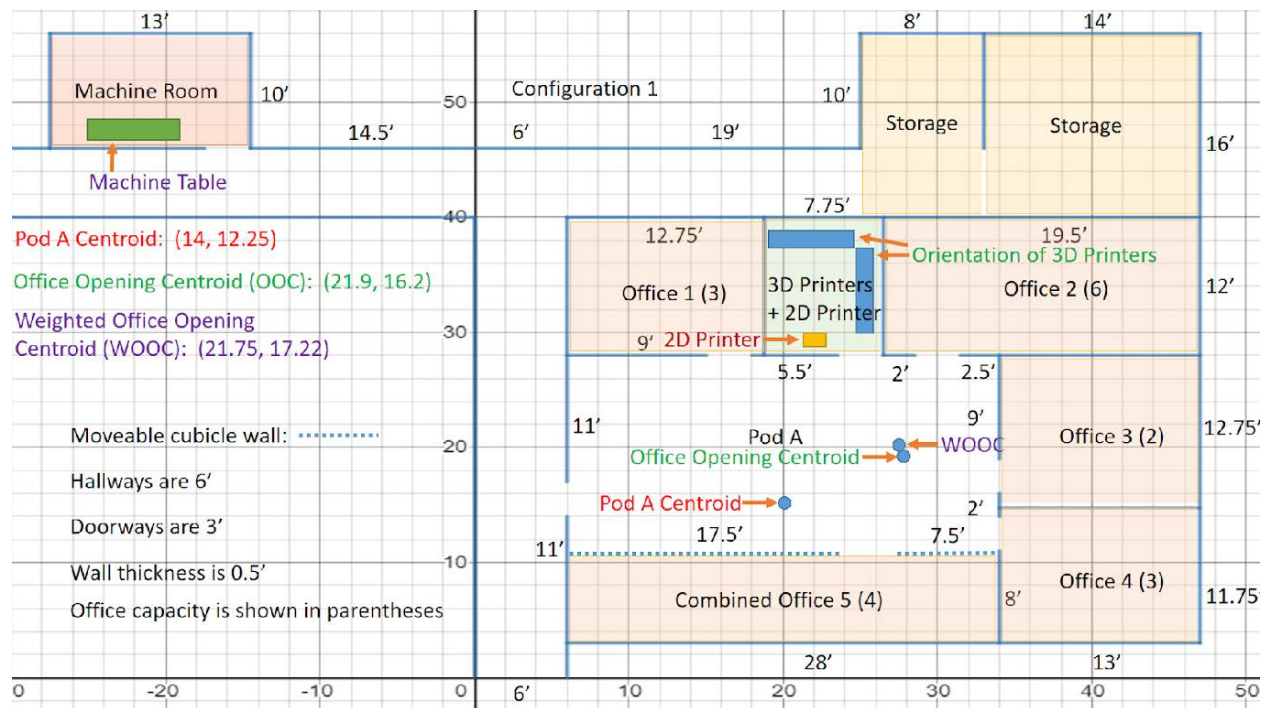


Figure 25. Combined Office Layout Concept.

The core motivation for proposing this layout is to capture notable gains in the quantitative metrics that are used to evaluate the suitability of the layout while also not significantly infringing upon the qualitative considerations as expressed by the governing stakeholders of the system at hand. The Lean principle described as “respect for people” is captured by allowing for some

amount of private office space while also making an attempt to reduce wasted movements. In this case, the wasted movements are measured by the travel distance required to reach a particular workstation when travelling from an office. For the purpose of analysis, the layout will be assessed on the basis of qualitative and quantitative criteria similar to the evaluation utilized to examine the previous two alternative layouts. The required changes needed to fully implement the concept of the combined office for the system under consideration are shown in the list consolidated provided below.

- Move the 3D printer into the initial state definition for Storage Room 2 (this becomes the 3D printer room)
- Move the items in Storage Room 2 into the initial state definition for the 3D printer room (this turns into storage space)
- Arrange the moveable cubicle walls in the orientation shown in the figure
- Move the 2D printer into the initial state definition for Storage Room 2 (becomes the 3D printer room) with the 3D printers

At this point, the layout under consideration will be analyzed under the qualitative measures proposed and emphasized by the governing stakeholders of the system at hand. This involves examining the suggested layout under the context of each qualitative consideration. Ultimately, the considerations contained under the domain of this qualitative analysis cannot be quantified or rigidly ranked. As such, judgment will be applied in accordance with the perception of the system under the circumstances proposed in a given layout alternative. The overall discussion and commentary shown below describes the qualitative implications of this particular layout alternative.

Qualitative Measures:

In total, this layout will be assessed on the basis of five qualitative considerations in order to determine the suitability of the layout in regards to the stakeholder preferences which were captured. These considerations consist of cost of implementation, mechanical flexibility, transparency of the space, worker privacy, and overall change from the current state. The results from assessing the proposed layout alternative are depicted in the figure shown in the space provided below.

Layout Qualitative Factors	Combined Office Layout
Cost of Implementation	Excellent
Mechanical Flexibility	Minimal Change
Transparency of the Space	Minimal Change
Worker Privacy	Minimal Change
Overall Change from the Current State	Minimal Change

Figure 26. Qualitative Results for the Combined Office Layout.

Given this establishment, the proposed layout will be analyzed from a quantitative perspective. While there are a number of quantitative approaches that may be applied in order to assess the suitability of a particular layout, the key metrics that were distinguished as critical by the governing stakeholders will be examined in order to form a quantitative foundation of analysis in regards to the associated alternative layout. The results of this quantitative analysis are depicted in the following section.

Quantitative Measures:

To begin, the previous quantitative techniques discussed were evaluated under the context of this particular layout. These assessments are a measure of distance, time, and cost related to non-value-added activities. Ultimately, these calculations are summed to yield a value which represents the total daily cost of walking. The results were then compiled and presented in the figure shown below.

Combined Office:	Average Distance	Average RT Time	Average RT Cost (\$)	Daily Cost of Walking Time (\$)
3D Printers (41 accesses/day)	9.31'	4.2 s	0.014	0.574
Storage (20 accesses/day)	64.44'	29.09 s	0.096966667	1.939333334
2D Printer (16 accesses/day)	9.31'	4.2 s	0.014	0.224
Machines (3 accesses/day)	66.55'	30.05 s	0.100150489	0.300451466
			Total Daily Walking Cost (\$): 3.037784806	

Figure 27. Quantitative Results for the Combined Office Layout.

At this point, it is desired to analyze the layout on the basis of an effectiveness calculation. This calculation will be assessed as a multiplication of the distance from each office to each workstation and the number of times that the trip is expected to be travelled in any given work

day. The summation of these effectiveness values would yield the expected distance travelled by all employees (between offices and workstations) in a regular work day of operations. In this system, the data collection phased showed that travelling from one workstation to another was very rare. Thus, trips were only conducted between an office and a workstation. Due to this phenomenon, the traditional representation of the effectiveness calculations will need to be altered to account for this fact. As such, the beginning distances chart is depicted in the figure shown in the space provided below.

Combined Office Layout	3D Printers	Storage	2D Printer	Machines
Weighted Office Opening Centroid (WOOC)	9.31'	64.44'	9.31'	66.55'

Figure 28. Average Distances from a Desk to a Workstation.

Given the table shown in the figure above, the total effectiveness of the layout will be obtained. Typically, the workstations which have the highest number of accesses per day or the longest distance travelled with contribute the most to the final value for effectiveness due to the method by which effectiveness is calculation. The figure below depicts the final calculations pertaining to effectiveness for this particular layout.

To → From ↓	3D Printers	Storage	2D Printer	Machines	Total
Weighted Office Opening Centroid (WOOC)	9.31 · 41	64.44 · 20	9.31 · 16	66.55 · 3	2019.12
				Total Effectiveness: 2019.12	

Figure 29. Effectiveness Assessment of the Combined Office Layout.

At this stage of the quantitative analysis pertaining to the layout at hand, simulation will be applied to assess the layout. The primary focal point of the application of simulation is to gain an understanding of the variability surrounding the travel times within the layout overall. To this point, travel time has been regarded as fixed and deterministic. Realistically, this is not the case. Employees have subtly varying walking speeds that should be accounted for. Due to the low volume in demand for the various work stations, the length of the queues within the system are typically zero (or a length of one in a rare case). Theoretical calculations under the domain of queueing theory verify this observable phenomenon. Simio was selected for application on this layout. In the simulation model, walking speed was modeled (involving $D = R \cdot T$) as a normally distributed (in the form of $X \sim N(\mu, \sigma^2)$) random variable in the generalized form of $R \sim N(4.43, 0.443^2)$. The figure below depicts the associated Simio simulation model for the layout under consideration.

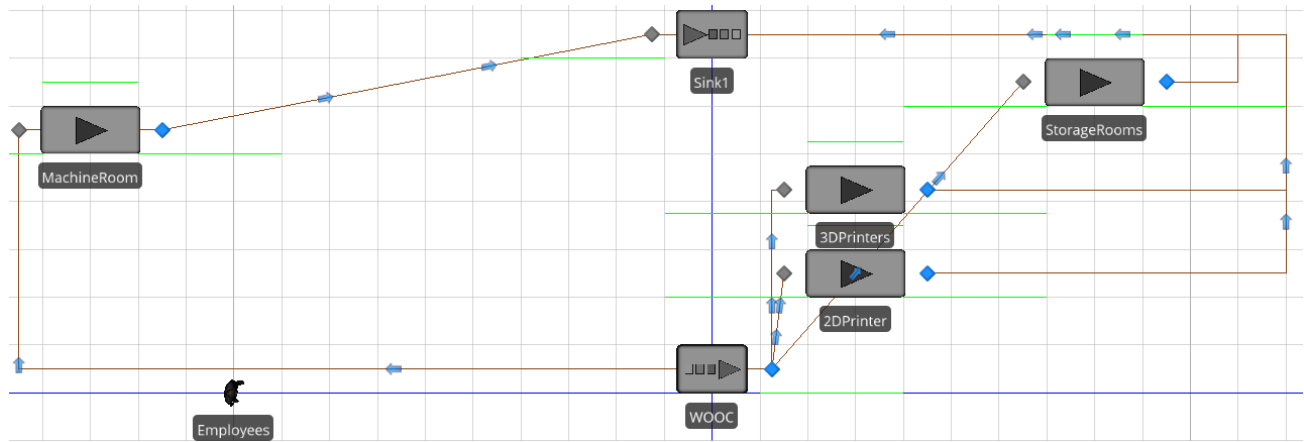


Figure 30. Simio Simulation Model for the Combined Office Layout.

Based on this model, an experiment was constructed using a total of 300 replications of the system, where each replication analyzed 2 hours of operational data. The mean time for a roundtrip (weighted based on the demand for each workstation) according to the simulation model was 11.26489 seconds with a standard deviation of 1.276544 seconds. Hand calculations show that the expected value (underlying mean) of this time is about 11.39458239 seconds. The figure below depicts the histogram and the associated box plot for the response statistic defined to be the average roundtrip time.

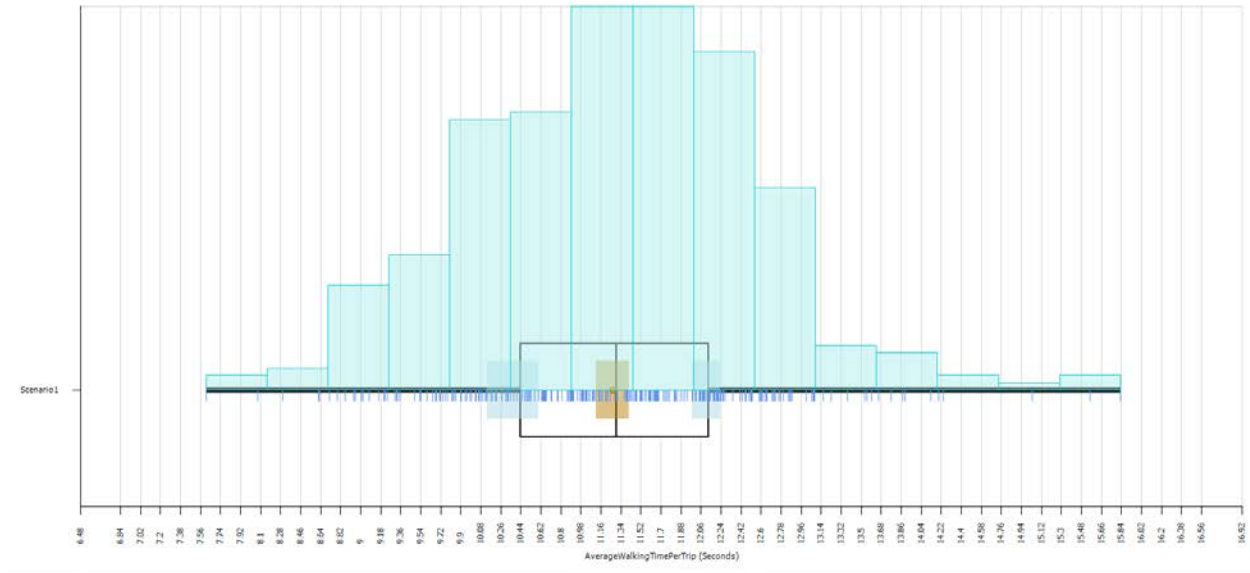


Figure 31. Histogram of the Average Walking Time Per Trip Statistic (n = 300).

At this point, it is desired to assess the validity of the simulation model by checking the consistency between the sampled data and the hand calculations of the theoretical mean value. This may be analyzed through the application of a hypothesis test. A typical hypothesis test for large samples will be applied to make this determination. In this case, the value for σ is not known, so the standard deviation, s , will be substituted. The resulting calculations of the hypothesis test are provided below.

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0$$

$$z = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}} \quad (s \text{ may substitute for } \sigma)$$

$$H_0: \mu = 11.39458239$$

$$H_1: \mu \neq 11.39458239$$

$$z = \frac{11.26489 - 11.39458239}{\frac{1.276544}{\sqrt{300}}} = -1.016$$

$$P - value \approx 0.3096$$

The results of the hypothesis test above indicate that the null hypothesis cannot be rejected when using a level of significance such as $\alpha = 0.05$. Thus, it is reasonable to conclude that the two mean values are equivalent within 95% confidence. Therefore, the simulation model constructed provides an accurate representation of the alternative layout proposed on the basis of this quantitative information. Anomalies are unlikely to arise due to random variation over the course of 300 replications lasting 2 hours each.

Next, this layout will be analyzed to assess the proportion of task operation time that should be generally expected under this layout. This will be calculated by dividing the operation time by the total time. In this case, the total time is defined as the sum of the operation time and the roundtrip travel time to reach the location of a workstation. The results of these calculations are provided in the figure shown below.

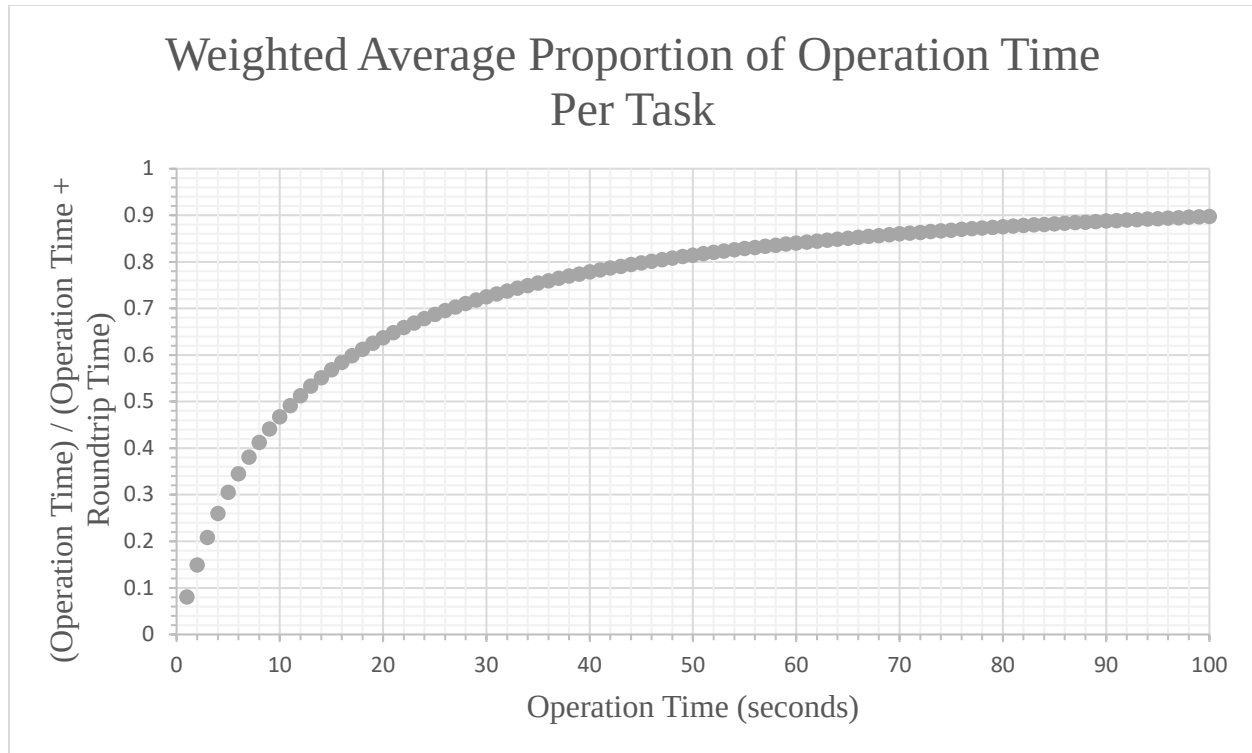


Figure 32. Weighted Average Proportion of Operation Time per Task.

This concludes the quantitative analysis procedures which will be performed on this particular layout. The insights gained from the manipulation of raw data, the Simio simulation model, and the expected proportion of operation time will be used as the foundation of analysis for the layout. Furthermore, the various alternative layouts will ultimately be compared in these regards in order to determine the layout which would best address the governing stakeholder needs as previously expressed.

5. Select the Criteria to Determine the Best Alternative:

The evaluation of each layout is analyzed relative to the initial state of the system. Thus, the associated benefits and detriments introduced by a particular alternative layout will be the

primary focal point. This perspective is utilized in order to provide an enhanced understanding of the types of benefits offered by a particular layout.

Overall, the governing stakeholders for the system expressed several objective criteria that will be used to assess each layout. These preferences were captured through reviewing strategic documents from the company and conducting interviews. However, the stakeholders did not place an equal level of emphasis on each criterion that was expressed. From a broad perspective, the stakeholders wish to minimize the qualitative drawbacks that a new layout would introduce while maximizing the quantitative gains.

In order to formulate an analysis which suggests the best layout based upon the stakeholder criteria, the specific criteria for comparison purposes were collected and compiled. The weighting of these factors will become a central focal point of discussion in the following section. The resulting list is provided below.

- Minimal distance between an office (or private office space) and a workstation
- Minimal time required to reach each workstation from an office
- Minimal cost associated with the non-value-added activity of walking from an office to a workstation
- No infringements on the governing constraints of the system (discussed in Section #3 on Assembling Relevant Data)
- Maximum performance on all qualitative factors

Given the objectives and the associated constraints of the system, it is important to acknowledge that the perfect layout does not exist. However, the proper modelling and analysis techniques may imply that a particular alternative is superior to another when striving to satisfy

the stakeholder's preferences. While a perfect layout would be one with no non-value-added activities and perfect performance on all qualitative factors, this type of solution does not lie within the realm of feasibility.

6. Construct a Model:

The evaluation of the various layout alternatives progressed on the basis of examining both quantitative and qualitative data in an independent fashion. The primary focal point of analysis is placed on comparing the quantitative and qualitative performance of the various alternative layouts to the initial state. Thus, the benefits and the drawbacks associated with each layout will be assessed relative to the initial state. The figure below provides a summary of the qualitative performance of each layout.

Layout Qualitative Factors	Open Office Layout	Systems Furniture Layout	Combined Office Layout
Cost of Implementation	Good	Poor	Excellent
Mechanical Flexibility	Excellent	Poor	Minimal Change
Transparency of the Space	Excellent	Poor	Minimal Change
Worker Privacy	Inferior	Excellent	Minimal Change
Overall Change from the Current State	Most	Moderate	Minimal Change

Figure 33. Qualitative Results Relative to the Initial State.

At this point, each qualitative factor was considered individually under the context of the three alternative layouts. A value of “Best,” “Moderate,” or “Worst” was assigned to each layout based on how well a particular layout performed on a qualitative factor relative to other layouts. Thus, the layout which performed the best on a certain qualitative factor would be designated as the “Best” layout for addressing this factor. For all of the qualitative factors under consideration,

this analysis procedure seems rather straightforward. However, the “Transparency of the Space” factor introduces a fascinating principle known as the transparency paradox. Studies have shown that higher levels of workplace transparency yield negative impacts on the overall working environment and culture [4]. Due to findings such as these, this study will consider higher levels of transparency to be less desirable than lower levels of transparency. For this reason, the labelling of “Best,” “Moderate,” and “Worst” for this qualitative factor are reversed from the convention used on the other factors. The resulting table depicting the outcomes of this labelling are shown below.

Layout Qualitative Factors	Open Office Layout	Systems Furniture Layout	Combined Office Layout
Cost of Implementation	Moderate	Worst	Best
Mechanical Flexibility	Best	Worst	Moderate
Transparency of the Space *	Worst	Best	Moderate
Worker Privacy	Worst	Best	Moderate
Overall Change from the Current State	Worst	Moderate	Best

Figure 34. Comparison of Qualitative Factors (* Transparency Paradox).

For comparison and weighting purposes, a unitless value was assigned to each “Best,” “Moderate,” and “Worst” category. In accordance with the circumstances of the project at hand, corresponding values of 1, 0, and -1 respectively were assigned. The figure below depicts these unitless value assignments.

Weightings on Qualitative Results		
Best	Moderate	Worst
1	0	-1

Figure 35. Quantitative Weights Place on Qualitative Results.

At this point, it is important to note that each qualitative factor should not carry the same weight due to the objectives of the study. A specific weight was assigned to each qualitative factor in accordance with the employee interviews conducted, the type of work currently performed and expected to be performed in the future, the findings of the literature review, and the strategic vision of the UAH SMAP Center. The figure below depicts the weights associated with each factor along with additional computations.

Layout Qualitative Factors	Value Weighting	Open Office Layout	Systems Furniture Layout	Combined Office Layout
Cost of Implementation	0.2	0	-0.2	0.2
Mechanical Flexibility	0.4	0.4	-0.4	0
Transparency of the Space	1	-1	1	0
Worker Privacy	0.4	-0.4	0.4	0
Overall Change from the Current State	1	-1	0	1
Total	3	-2	0.8	1.2

Figure 36. Weighting of Qualitative Factors and Evaluation.

As depicted in the figure shown above, the most preferred layout (among the three alternative layouts considered) on the basis of qualitative considerations is the combined office

layout. This conclusion was reached through an assessment of the governing stakeholder preferences along with the findings of the literature review. The next section describes (in higher detail) the conclusions which may be drawn about the three alternative layouts from a quantitative analysis perspective.

7. Predict the Outcomes for Each Alternative:

At this point, the quantitative results of each layout will be compared in a similar manner to the previous comparison of the qualitative factors. Furthermore, the initial state will be considered in order to identify specific areas where a certain alternative layout performs better in a particular quantitative metric. The resulting analysis of the alternative layouts are depicted below.

Open Office Layout:

The noteworthy strength of the open office lies in the quantitative metrics. As such, there are numerous notable gains provided by this alternative layout from a quantitative perspective. Overall, it is estimated that adopting this layout would reduce daily costs by 63.68% and would save \$3.98 per day which translates to \$1,452.60 per year. Furthermore, the calculated effectiveness of this layout was shown to be 1506.89. The figure below depicts the quantitative results of this layout.

Open Office:	Reduced Distance	Reduced RT Time	Reduced RT Cost (\$)	Reduced Daily Walking Cost (\$)
3D Printers (41 accesses/day)	53.66'	24.23 s	0.080748865	3.310703454
Storage (20 accesses/day)	19.32'	8.72 s	0.029069458	0.581389161
2D Printer (16 accesses/day)	1.36'	0.61 s	0.002052793	0.032844682
Machines (3 accesses/day)	11.53'	5.21 s	0.017353674	0.052061023
			Reduction in Total Daily Walking Cost (\$): 3.97699832	

Figure 37. Cost Reductions from Implementing the Open Office Layout.

The primary concerns surrounding an open office layout are centered upon the qualitative factors expressed by the governing stakeholders. The open office can be intolerant to work that is individualistic rather than collaborative. While employees housed within the system perform a

mix of both individualistic and collaborative work, the open office would only offer a limited amount of compatibility with this arrangement. However, the quantitative gains may be appealing enough to adopt an open office layout under the circumstance where the nature of the work shifts further towards collaboration.

Systems Furniture Layout:

The promising feature of the systems furniture layout is the allowance of personal office space while still offering improvements on several of the quantitative metrics. In contrast to the open office layout, the systems furniture layout does not place a strong emphasis on solely quantitative or solely qualitative factors. The layout strategically targets improvements in both categories. Overall, it is estimated that adopting this layout would reduce daily costs by 68.07% and would save \$4.25 per day which translates to \$1,553.42 per year. Furthermore, the calculated effectiveness of this layout was shown to be 1323.6. The figure below depicts the quantitative results of this layout.

Systems Furniture:	Reduced Distance	Reduced RT Time	Reduced RT Cost (\$)	Reduced Daily Walking Cost (\$)
3D Printers (41 accesses/day)	59.62'	26.92 s	0.089725331	3.678738574
Storage (20 accesses/day)	19.39'	8.75 s	0.029178993	0.583579857
2D Printer (16 accesses/day)	(-) 1.5'	(-) 0.68 s	(-) 0.002261494	(-) 0.036183901
Machines (3 accesses/day)	5.96'	2.69 s	0.008969541	0.026908623
			Reduction in Total Daily Walking Cost (\$): 4.253043153	

Figure 38. Cost Reductions from Implementing the Systems Furniture Layout.

While the systems furniture layout offers some notable gains in the quantitative metrics contained under the scope of this study, only minor gains are offered in regards to the qualitative factors. However, this analysis shows that the systems furniture layout performs better from a

quantitative perspective than the open office layout does. Fundamentally speaking, this is a unique gain because minimal drawbacks appear to be incurred from capturing these types of improvements.

Combined Office Layout:

The quantitative outcomes of the combined office layout depicted improvements in a number of areas. However, these improvements are largely centered around two workstations (access to the 3D printers and the 2D printer). While the gains made on the quantitative metrics for this layout are milder than the gains displayed by the other two layouts, the combined office is clearly superior when the qualitative factors are examined. Overall, it is estimated that adopting this layout would reduce daily costs by 51.36% and would save \$3.21 per day which translates to \$1,171.47 per year. Furthermore, the calculated effectiveness of this layout was shown to be 2019.12. The figure below depicts the quantitative results of this layout.

Combined Office:	Reduced Distance	Reduced RT Time	Reduced RT Cost (\$)	Reduced Daily Walking Cost (\$)
3D Printers (41 accesses/day)	62.51'	28.23 s	0.094085	3.857485
Storage (20 accesses/day)	(-) 27.56'	(-) 12.44 s	(-) 0.041464667	(-) 0.82929334
2D Printer (16 accesses/day)	7.66'	3.46 s	0.011541	0.184656
Machines (3 accesses/day)	(-) 1.23'	(-) 0.56 s	(-) 0.001847489	(-) 0.005542466
			Reduction in Total Daily Walking Cost (\$): 3.207305194	

Figure 39. Cost Reductions from Implementing the Combined Office Layout.

The gains offered by the combined office layout are primary qualitative in nature, but this layout also captures key gains when considering the quantitative factors. From a broad perspective, the appropriate layout seeks to obtain maximal gains in both categories. However,

considering that this is not necessarily feasible, it is desirable to strike a healthy balance in order to best satisfy the governing stakeholder preferences and objectives. The following sections describe these recommendations in further detail.

8. Choose the Best Alternative:

After examining the results of each layout, the findings were expressed in a comparable manner in order to select the proper alternative layout. The figures below depict the savings incurred by each layout on a daily and yearly basis. Additionally, these values are expressed as a percentage efficiency relative to the current state. A figure is also provided in order to express the effectiveness of each layout in a comparable fashion.

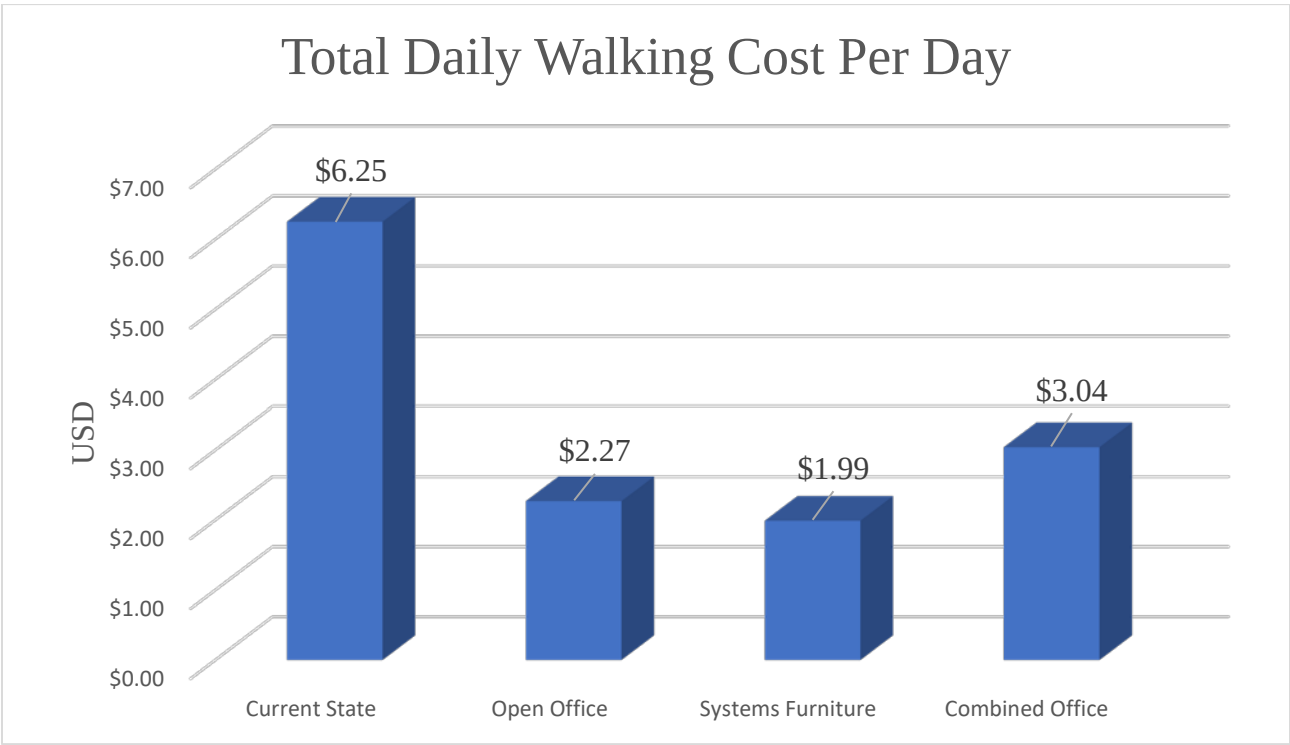


Figure 40. Daily Labor Cost Incurred due to Employees Walking to Workstations.

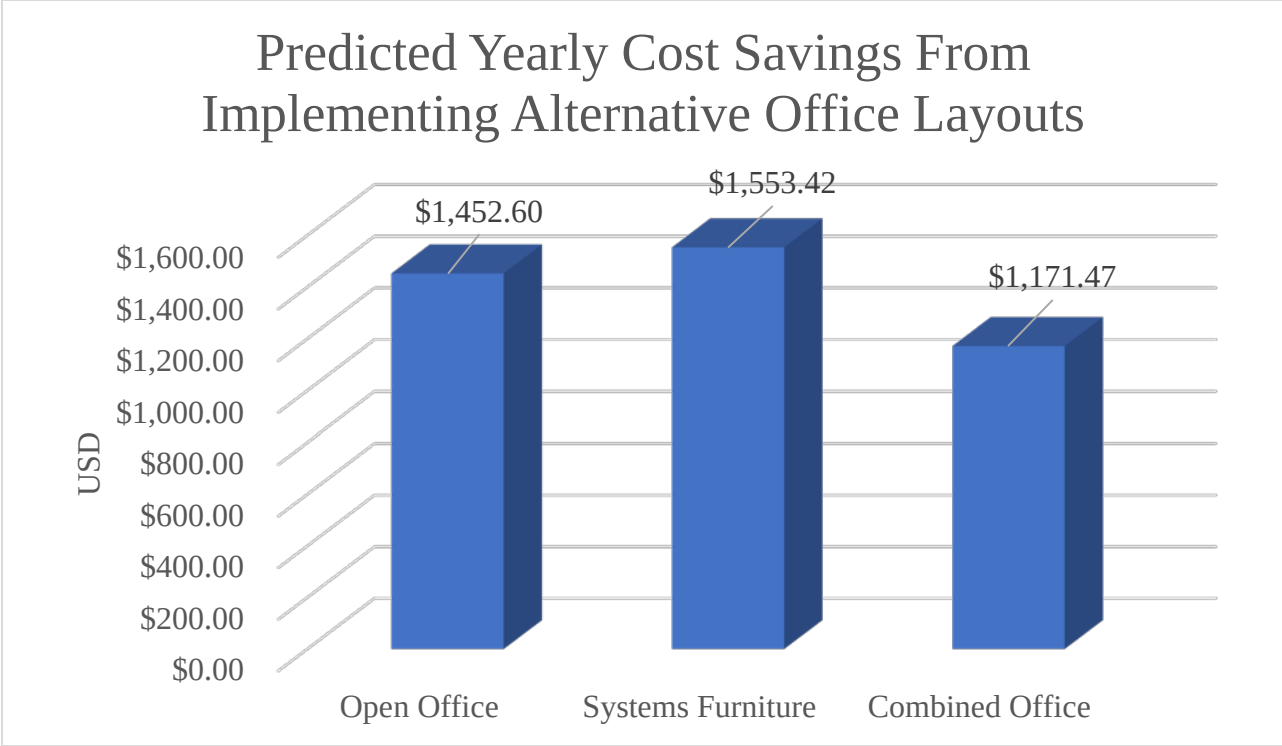


Figure 41. Yearly Savings in Walking Cost for Each Alternative Layout.

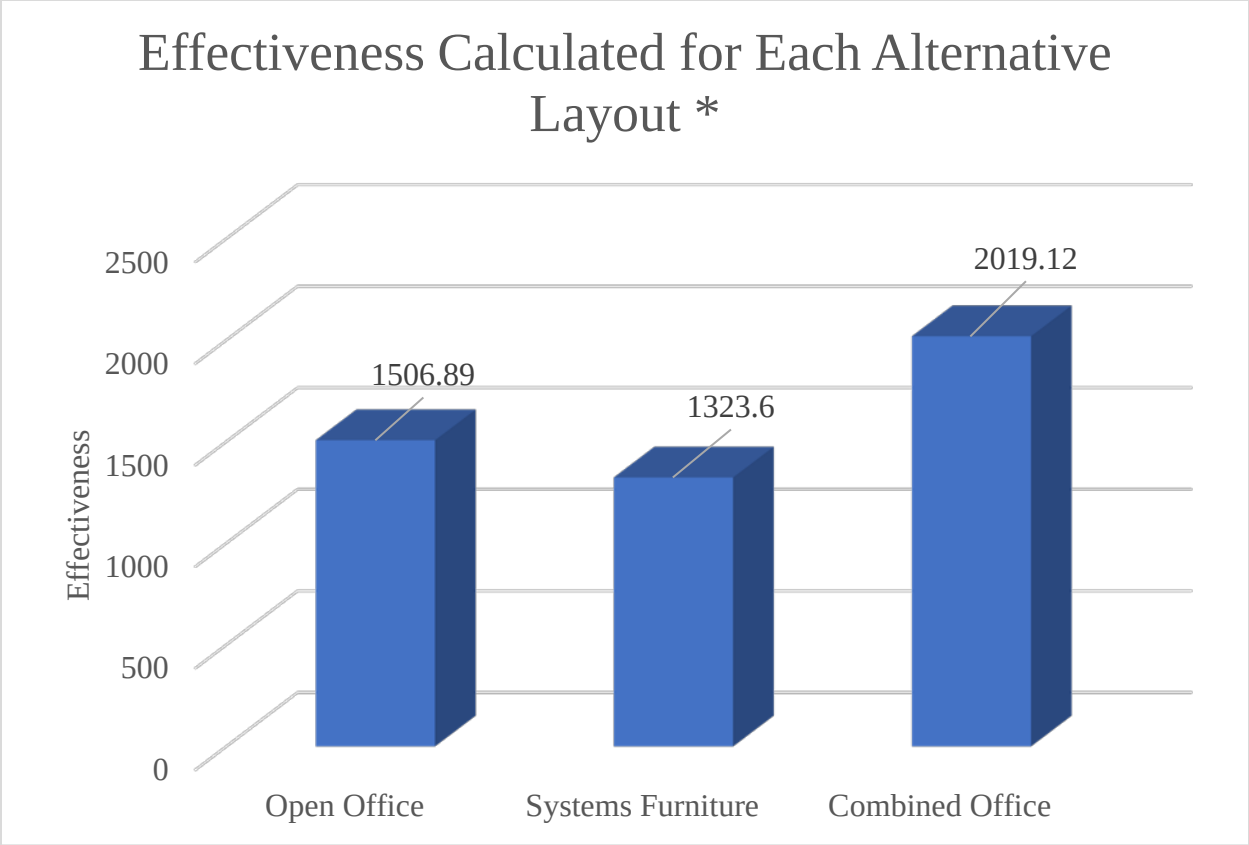


Figure 42. Effectiveness of Each Alternative Layout (* Minimization Objective).

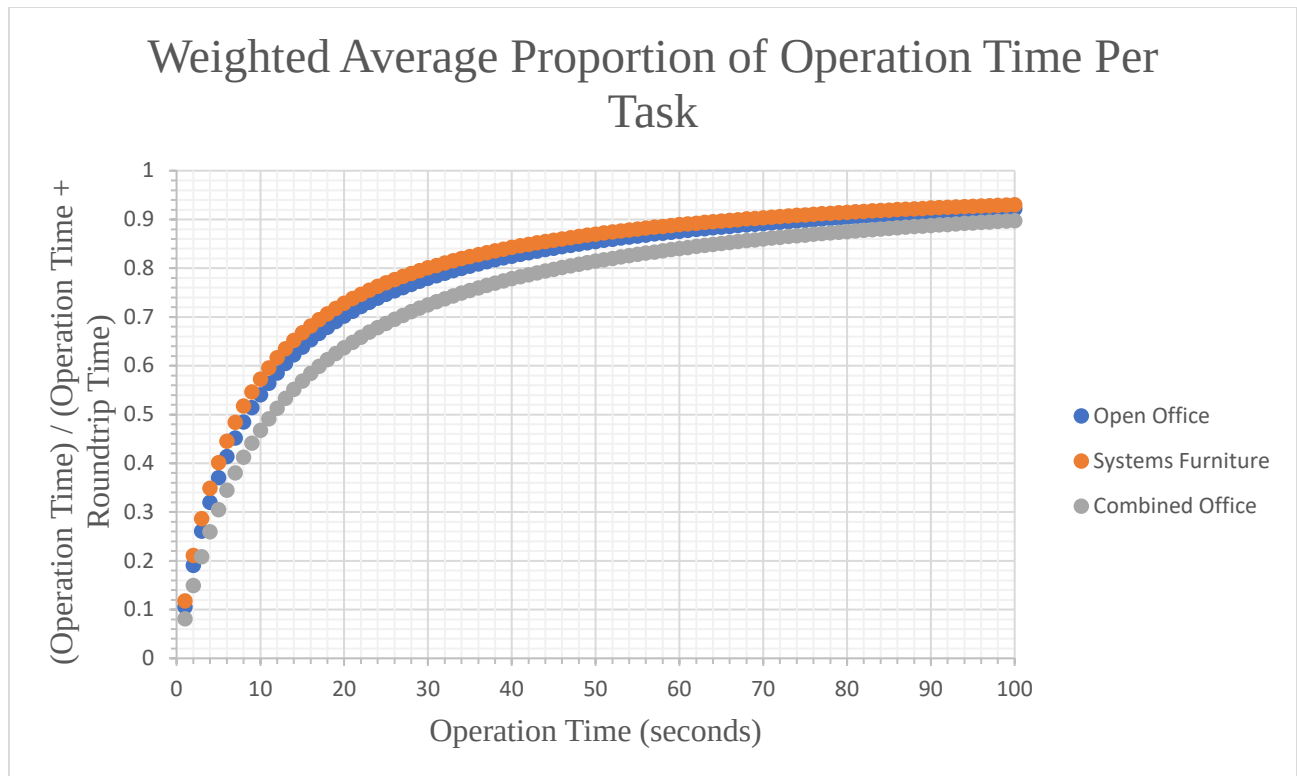


Figure 43. Weighted Average Proportion of Operation Time Per Task for Each Layout.

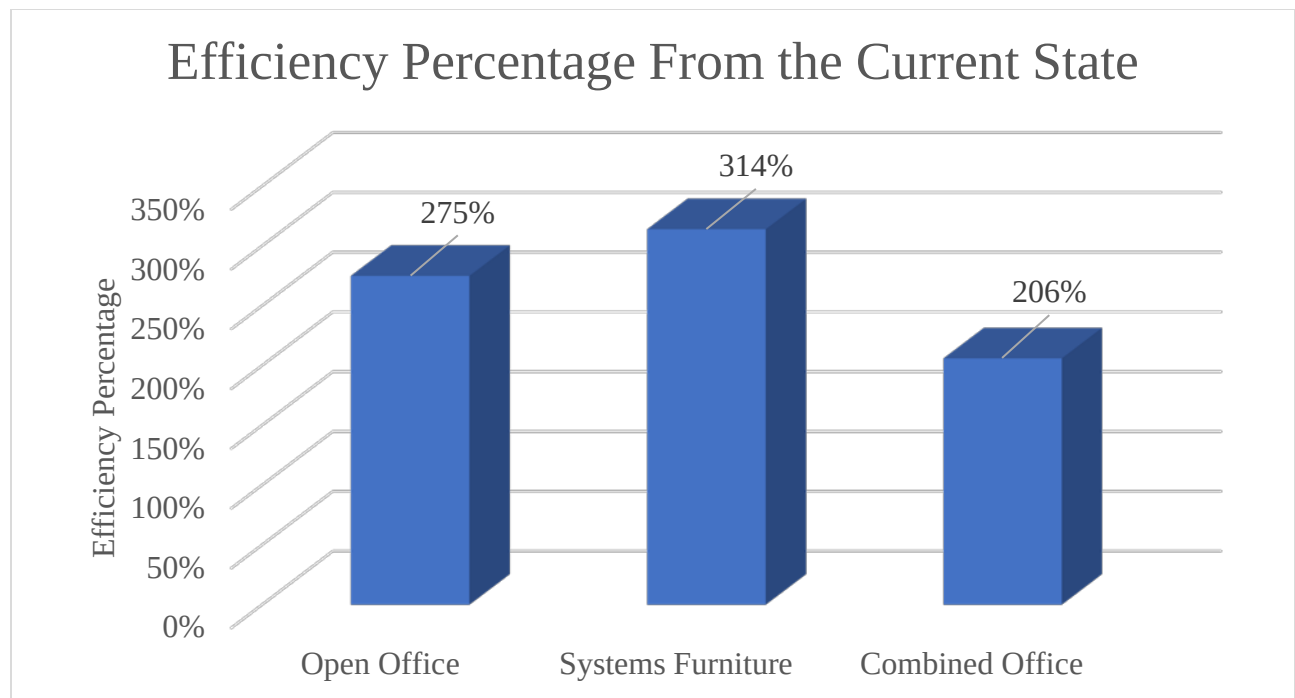


Figure 44. Efficiency Percentage Associated with Each Configuration.

The results depicted in the figures above indicate that any one of the alternative layouts would double the benefits associated with the quantitative metrics. This is evident by the efficiency percentage expressed in the last figure shown. The lowest efficiency percentage is 206% and the highest is 314%. These numbers show that excellent gains could be made by implementing any one of the alternative layouts. However, the qualitative results shown in the figure below depict a much different conclusion in regards to the qualitative factors.

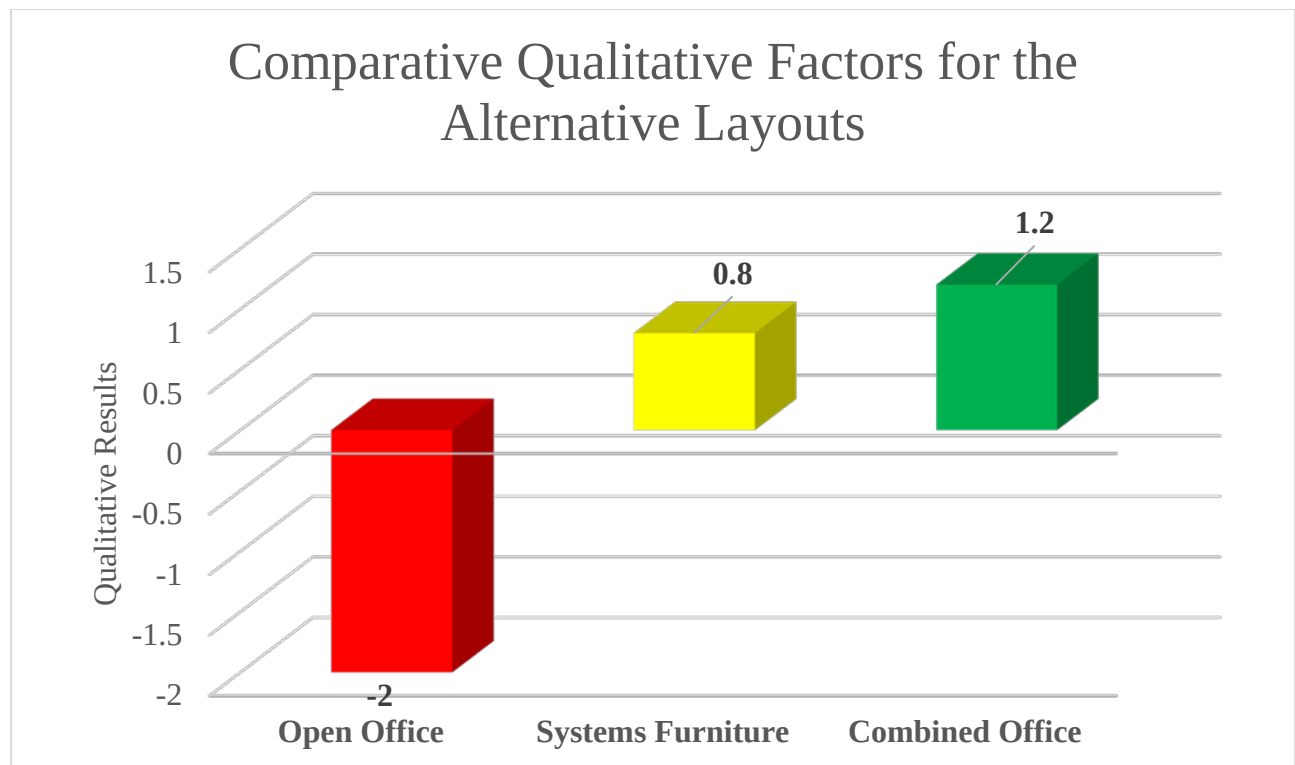


Figure 45. Qualitative Results for Each Configuration.

As shown in the figure above, the combined office is largely superior on the basis of the qualitative factors taken into consideration. As expected, the open office suffers in this regard, but it succeeds in capturing large magnitudes of quantitative gains. The systems furniture layout is unique because it captures notable gains in both the qualitative and the quantitative categories. Beyond the numerical results, there are several situational circumstances that would need to be addressed upon implementing any of the alternative layouts considered under the scope of this study. The section below describes the recommendations of the study at the conclusion of the analysis procedures.

Recommendations:

Upon considering the results of the study, the recommended alternative layout to best address the governing stakeholder needs is the combined office layout. This layout is recommended because of the significant gains made on the basis of the qualitative factors. Likewise, this layout still produces substantial gains on the basis of the quantitative metrics. The combined office layout is unique because no single qualitative factor was negatively impacted by the introduction of this layout. While marginal gains in the quantitative metrics may be found through the exploration of other layouts, these alternatives come at the cost of critical qualitative factors. It is important to note that as the magnitude of change relative to the initial state rises, the unexpected outcomes produced from the variables which were not captured under the scope of the study increase. Thus, the combined office represents a conservative approach to obtaining significant gains. When implementing this layout, there are a wide variety of small adaptations to the layout which may offer the capability to capture incremental gains in both quantitative and qualitative factors. In this regard, the combined office layout possesses an element of flexibility in order to adapt to changing circumstances. Additionally, it is important to closely monitor the results of the combined office in order to detect the occurrence of any unexpected outcomes. When detected, the combined office may hold enough flexibility in order to bypass these potential drawbacks.

9. Audit the Results:

A review of the ignored variables of this study is recommended before pursuing the full implementation of this layout. The findings from the data collection phase, employee interviews,

strategic documents, and the literature reviews indicate that the changes of encountering these unexpected outcomes are minimal, but assessing this possibility before and after implementation occurs is nevertheless critical. These steps should be taken to ensure the success of the implementation of the combined office layout.

Similarly, further data collection activities may be conducted in order to determine if the underlying assumptions of the study at hand are indeed reasonable. This study takes a rigorous and critical stance in regards to assumptions made. The assumptions stated in this study are unlikely to cause any consequential blowback in the implementation stage. However, systems are subject to a natural level of change over time as the work culture evolves along with the type of the work performed. These types of changes and gradual transitions should be monitored to ensure that the circumstances of the system do not invalidate the findings of the study. The data collection phase of this study concluded that long-term levels of change (which would impact the recommendations) are unlikely to arise. However, this critical factor should never be ruled out entirely.

One important aspect of the implementation phase should involve continual feedback from employees and management. This begins with a declaration of approval from both employees and management prior to engaging in the implementation of the combined office layout. Additionally, a survey should be developed in order to assess several of the qualitative factors while implementation is occurring along with the long-term impacts of the new layout. Throughout this process, it is important to allow employees and management to voice any potential concerns. This closely aligns with the critical Toyota Production System principle commonly known as “Respect for People.”

One potential change that should be considered when conducting studies of this type is the adoption of the techniques and methodologies utilized by Frederick Winslow Taylor in his time studies. Instead of approximating travel times using a model, travel times can be sampled from every day operations. However, this data can be difficult to capture due to the Hawthorne Effect. People often change their behaviors when they know that they are being watched or recorded in any fashion. Overall, this is merely one of the possible considerations for improvement when conducting collecting this type of data.

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